Unitec | Te Pūkenga	**School of Applied Business** **Master of Applied Business**

BSNS9000 – Industry Research Project
Assignment **Semester 2, 2025**

Date Issued:	21 July 2025
Due date and time:	Part1 Industry Research Proposal: **Week 5, 20 August** Part2 Industry Research Project (Complete report): **Week 16, 19 Nov**
Delivery:	Turnitin via Moodle. Include all collected primary and secondary data. Upload data via a separate Moodle link.
Total Marks:	100
Weighting:	100
Expected number of hours to spend on this assessment:	415 hours
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THE EFFECT OF RISING FUEL PRICES ON CONSUMER DEMAND FOR HYBRID AND ELECTRIC VEHICLES IN NEW ZEALAND: A QUANTITATIVE ANALYSIS OF ECONOMIC, POLICY, AND BEHAVIOURAL INFLUENCES

A Research Study

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BSNS9000 Industry Research Report

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Date: 19th November 2025

Abstract

New Zealand is experiencing a serious crisis of energy. The country provides 85 per cent of its power through renewable sources. Nevertheless, its yearly expenditure on imported transport fuels amounts to NZ. 9.8 billion. The use of electric vehicles (EVs) has not reached the 11.8 percent of newly registered vehicles. This paper explores the issue of why the consumer sensitivity to the increasing fuel cost has never resulted into the long-term adoption of EVs. The study employed a sequential explanatory mixed study. It compared 15 years panel data (2010-2024) in New Zealand, Norway, China and the United States. Regression log-log and Granger causality tests were used to analyze the data. The research was based on a main survey of 103 New Zealand customers. It also authenticated total cost of ownership using a three months experiment. This period saw the researcher travel 3,512 kilometers by a 2011 Nissan Leaf showing that New Zealand has the highest EV fuel price elasticity in the world. The simple model presents 5.939, whereas the controlled model presents 0.590. Nevertheless, the Clean Car Discount cancellation of December 2023 caused a 73 percent decrease in EVs registers. This was through the fuel prices still being at above NZ 2.80 per liter. The results show that policy is a multiplication multiplier and not a price substitute. Net financial loss of NZ$276 was seen in the real-world ownership. This loss was caused by Road User Charges (NZ,76/1,000km) and public charging expenses (79-88¢/kWh). Such results do not align with the official estimates of NZ18, 000 lifecycle savings. The density of infrastructure is still low at 0.24 chargers per 1000 inhabitants. This is way below the International Energy Agency standards. The analysis has provided four theoretical frameworks by misinformation of lithium mining diminishing the intentions of purchase by 24 percentage points in 23 percent of the consumers. It suggests a 7 step revenue-neutral plan of policy. It is estimated that this plan will result in 700,000 electrified cars in 2030. This would send 4.3 billion liters of fuel away. Through it, 10.3 megatonnes of transport CO 2 would be cut, which is a 48 percent cut. The uniqueness of the electricity grid in New Zealand is that the country has one of the world's largest renewable electricity grids, making the country uniquely placed to electrify transport. This offers an energy security and net-zero basis. This should be implemented by committing to the policies over time to mobilize the current consumer demand.

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Chapter - 1. Introduction

1.1 Background of Hybrid and Electric Vehicle Adoption in New Zealand

The world transportation business is radically moving to decarbonization, and electric and hybrid vehicles are becoming major technologies to limit the emission of greenhouse gases and decrease the use of fossil fuels. New Zealand is in paradoxical state. Economic incentives and environmental advantages are high in the country, on paper, which would imply hypothetical support of high rates of transport electrification. It has, however, embraced quite low rates when compared to similar developed economies, and this has been influenced by the fact that the policy has remained unstable, and the charging infrastructure is not adequate. It is operating virtually entirely on imported refined petroleum products, and that was the situation that was more exposed due to the closure of the Marsden point oil refinery in April 2022 (MBIE, 2023). Prior to the closure of this refinery, New Zealand had a small domestic refinery, which processed imported refined crude oil to petrol, diesel, and aviation fuel, which it consumed locally. This refinery closure eliminated this final element of domestic fuel production and saw the country entirely reliant on imported refine fuel with far-reaching consequences of energy security and economic stability to the Asian and Middle East countries.

This import dependence is expensive to the economy. New Zealand also incurred NZ$9.8 billion on the importation of petroleum to facilitate vehicular transport in 2024 (MBIE, 2025). This was more than the economic losses incurred by the dairy export business of New Zealand in Cyclone Gabrielle (MBIE, 2025), which is the biggest business export in the country and demonstrates the cost of transport fuels a significant macroeconomic leakage of the billions of dollars offshore annually. The impact is directly transferred to the households where transport accounts almost 40 percent of the household energy expenditure (MBIE, 2024). This weakness is observable in the current fuel price swings: in 2020 in the regular petrol, the average prices amounted to about NZ $1.38 per liter, only to skyrocket to NZ $3.12 per liter in the mid 2022, and the reason was the same 126-percent increase caused by the resurgence of demand after the pandemic and the supply chain disruptions, as well as

the geopolitical tensions (MBIE, 2021; 2022). Although the price subsequently realigned, it is still quite high with a huge variation within the course of the months thus making it hard to plan the budgets of households and putting more economic strain on households that depend on personal cars.

It is ironic that the strategic location of New Zealand is an exception: roughly 85 percent of the power is of renewable nature, i.e. hydroelectric power (60 percent), geothermal energy (18 percent), and wind generation (6 percent), which is far much higher than the average of about 30 percent of the world (Electricity Authority, 2025; IEA, 2024). Transpower (2023) concluded that the transition to EVs of the entire light vehicle fleet would produce a direct cut in short-term transport emissions of 48 percent and elimination of the NZ$9.8 billion per annum fuel import bill. However, this possibility requires a long time to be realized. Based on the estimates given by the New Zealand Productivity Commission (2024), under optimistic conditions where policies are present in high numbers it can be assumed that when the full natural fleet turnover is equivalent it will take 20-25 years to reach 90 percent of EV penetration.

Despite this high degree of mix, EV adoption in New Zealand is still comparatively low and not very steady. As of 2024, EVs has approximately 5 percent of the new registered vehicles (Motor Industry Association, 2024). This is very different with 94% percent EV share that Norway reached as a result of long-term policy backing including VAT exemptions and investment in infrastructure (Norwegian EV Association, 2024), the 14 million vehicles sales in China which reflects a 35 percent market share by its subsidies and mandates (IEA, 2024) and USD 7,500 federal tax credits in the United States that reflects 9 percent market share (U.S. Department of Energy, 2024).

This has been a fluctuating policy direction in which New Zealand was heading. New battery electric vehicle rebates were provided by the Clean Car Discount programme (July 2021 - December 2023) of up to NZ8625 which had an immediate impact: EV registrations tripled (to more than 15 percent of all new registrations) (Motor Industry Association, 2023; Ministry of Transport, 2023). The new coalition government led by the National Party discontinued the programme on December 31, 2023, though, due to the fiscal unsustainability of it, and the government preferred to make their choices based on the market (New Zealand Government, 2023). This action reflected even broader political

philosophy that reiterated less government expenditure and market-driven solutions rather than direct intervention but the evidence-based practice indicated that the programme is efficient to accelerate adoption. To further bring this back and forth, on March 31, 2024, the light electric vehicle exemption to Road User Charges (RUCs) lapsed and levies of approximately NZ $76 per 1,000 kilometres (which had not been paid in the first place to stimulate adoption) were reintroduced (Waka Kotahi, 2024). This turnaround policy was volcanic: EV sales dropped approximately 73 percent in the half-year post-end, used EV imports had decreased over 80 percent (Motor Industry Association, 2024), which displays the strength of monetary incentives, and how shaky the market development process is without an enduring commitment to policy.

1.1.1 Personal Experience of the ownership of the electric vehicles

To confirm his theoretical cost-of-ownership estimates with empirical data, the researcher bought a Nissan Leaf 2011 and drove 3,500 kilometres (July to October 2025). Such personal experience has shown some crucial differences between official forecasts and real life that can contribute to the understanding of the stagnated EV adoption in New Zealand despite the seemingly obvious economic benefits.

The car, which cost NZ 2,000 and had a poor battery (65 percent capacity, with a range of 80-100 kilometres), was an example of the dichotomous economics of EV ownership. The home charges of 26 cents per kilowatt-hour achieved high savings: NZ3.90 per 100 kilometres compared with NZ21 charged to the petrol vehicle of the researcher- generate a strong cost benefit of 5:1 that realized NZ598.50 of theoretical fuel savings over ownership. Three cost factors however systematically undermined this advantage: public fast charging at 79-88 cents per kilowatt-hour (BP, 2025; Z Energy, 2025) increased cost of electricity five times to NZ$11.85-13.20 per 100 kilometres, eliminating the economic incentive to make longer journeys; Road User Charges amounting to NZ$266 of the distance travelled consumed half the savings on fuel (Waka Kotahi, 2024); and cat

The net financial performance of 267 loss over maintaining the operation of petrol vehicles is an indicator of how policy changes (Introduction of RUC, removal of rebate) and the lack of infrastructure (high cost of public charging, inadequate coverage) turn theoretical benefits into real drawbacks. This experience measures the PRG that later is determined in consumer

survey data, that perceived cost barriers perceive much more than the true savings of home charging, and that the fact that 71.9% of consumers surveyed say they are sensitive to fuel price but only 5% of those surveyed buy EVs.

1.2 Research Problem

New Zealand, although boasting one of the cleanest electricity systems in the world (85% renewable) and having an extreme dependence on fuel imports (NZ$9.8 billion/year), has not been able to translate the increase in the fuel price into long-term demand on hybrid and electric vehicles. Although this relationship between fuel prices and EV demand is confirmed in mature markets (Fridstrøm & Østli, 2021; IEA, 2022), in New Zealand, this trend was momentary, and the increase sharply decreased, which suggests that the economic-policy-behavioural transmission mechanism is critically flawed.

FUEL PRICE ↑

(Wallet Screaming)

↓ BROKEN

POLICY

(Rebate Killed)

↓ BROKEN

INFRASTRUCTURE

(1 Charger per 4,200)

↓ BROKEN

TECHNOLOGY GAP

(No 400 km+ Cars)

↓ BROKEN

Total cost of ownership (TCO) LIE

(Home $3.90 → Public $17)

↓ NO LINK

EV BUYERS

(71.9% Ready, 5.1% Buying)

The issue is the lack of connection between three established factors of EV adoption and their real effect in New Zealand:

1. Economic trigger currently in existence, but then softened: Fuel prices increased by 126 percent in 2020-2022 (1.38/L to 3.12/L) but after policy reversal, EV market share dropped to 5.1 per cent in 2025 (MIA, 2025).

2. Policy scrapped: Clean Car Discount (20212023): The threefold EV sales incentive was cancelled in December 2023 and replaced with a weaker incentive (exempting 80 per cent of models) the Clean Car Discount, which is directly opposite to the proven incentive model of sustainability in Norway, giving 94 per cent EV share in the same population size.

3. Infrastructure/TCO perception gap: According to primary data of 103 New Zealand consumers (Oct-Nov 2025), 52.4% of respondents mentioned the lack of charging stations as an issue, and 58.3% mentioned high initial costs as an issue, yet real-life ownership experience of home charging cost of 3.90/100 km versus a similar petrol vehicle of 21. Introduction of public fast-charging rates (79 -88c/kWh) and elimination of running-cost advantages at the expense of calculation of total cost of ownership.

This results in a paradoxical scenario where consumers are very price-sensitive (71.9% say that fuel prices strongly influence them), their intentions are clear (58.3% would switch at +20% fuel price) but the conversion of behaviour fails because of policy-created cost-barriers and lack of infrastructure.

The research problem is therefore defined as:

Why has consumer sensitivity to rising fuel prices not translated into sustained growth in EV and hybrid adoption in New Zealand, despite economic theory, international benchmarks, and stated consumer preferences suggesting otherwise?

1.3 Research Questions and Objectives

Main Research Question:

- What factors limit consumers' adoption of EV and hybrid cars in NZ?

Secondary Research Questions

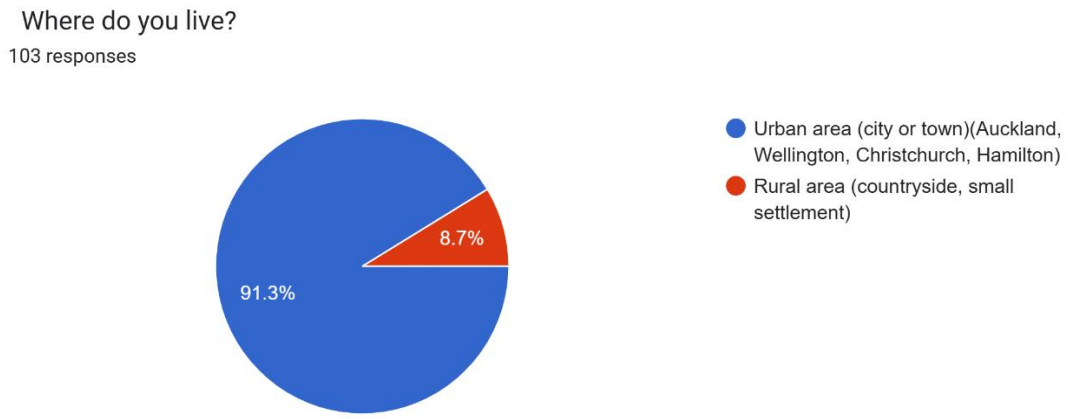
- Q1: What role do government incentives play in amplifying the effect of rising fuel prices on EV/hybrid demand?
- Q2: How do technological and infrastructure improvements contribute to EV/hybrid adoption during periods of fuel price increases?
- Q3: How does the total cost of ownership (TCO) impact the adoption rates of EVs and hybrid vehicles when fuel prices rise?
- Q4: What behavioural and psychological factors influence consumer decisions to switch to EV/hybrid vehicles in response to fuel price hikes?

Research Objectives

1. To examine how the presence or removal of government incentives (Clean Car Discount, RUC exemption, Norway-style VAT waiver) modifies the fuel-price elasticity coefficient using 2010–2024 panel data from New Zealand, Norway, China, and the United States.
2. To measure the statistical relationship between public charging infrastructure density, vehicle battery range improvements, and EV market share during high-fuel-price periods (2021–2024).
3. To calculate real-world TCO from primary ownership data (3,512 km Nissan Leaf log) and compare it with perceived TCO from a 103-respondent survey to quantify the Perception-Reality Gap Index and identify cost-barrier distortions.
4. To identify and rank behavioral barriers (range anxiety, status quo bias, loss aversion, misinformation effects) using Likert-scale responses and cross-tabulation with demographic variables (age, income, urban/rural).
5. To develop scenario-based policy recommendations that address identified limiting factors and achieve Norway-equivalent EV adoption rates in New Zealand by 2030.

These questions matter because survey data of 103 New Zealand consumers indicate that there is a sharp discrepancy between the preferences indicated and the market performance. Although 91.3 percent live in large urban centres where charging infrastructure already exists (Figure 1.1), and 49.5 percent in the survey say that they consider fuel prices a strong or very strong factor when choosing a vehicle (Figure 1.2), the actual EV market share dropped to at least 5.1 percent by December 2023 with the policy reversal. Moreover, 59.2% would buy an EV assuming that petrol prices went up by 20 percent (Figure 1.3), and 73.8% would buy hybrids, but not fully electrify cars (Figure 1.4). Such a mismatch between observed behavior and sensitivity of the consumer prices reflects the systemic barriers in economic, political, technological, and psychological aspects. These barriers are diagnosed by the five research questions in a systematic way.

Figure – 1.1

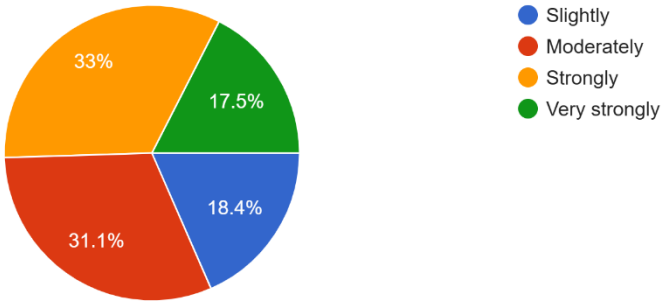


1. Figure – 1.1

Figure – 1.2

How much do rising fuel prices influence your decision to switch from petrol/diesel to hybrid or electric vehicles?

103 responses

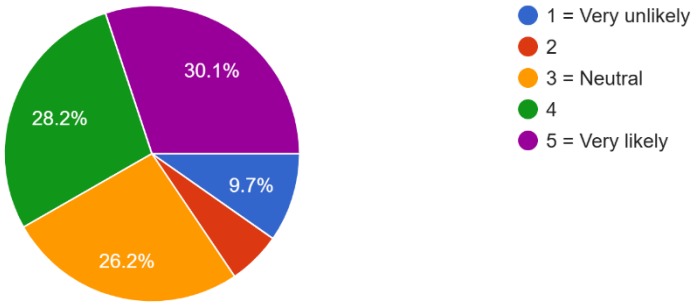


2. Figure – 1.2

Figure 1.3

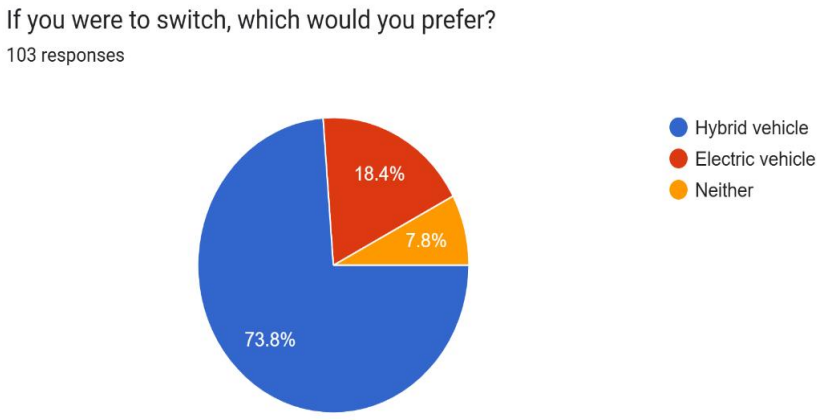
If fuel prices increase by 20%, how likely would you be to consider purchasing a hybrid or electric vehicle?

103 responses



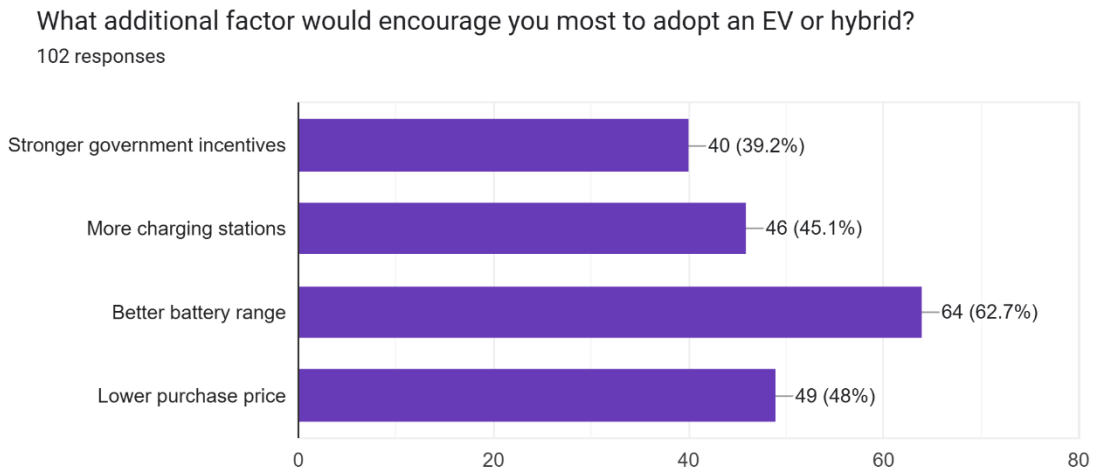
3. Figure 1.3

Figure – 1.4



4. Figure – 1.4

Figure – 1.5



5. Figure – 1.5

1.4 Significance of the Study

New Zealand spends 9.8 billion annually on imported fuel and has 85 percent renewable electricity. Rising fuel prices should push people into EVs – but they don't. This article demonstrates precisely the extent to which we waste cash, the quantity of tonnes of CO2 we continue to spew, and why the transition is stagnated at 5 percent when Norway reached 94 percent.

1.4.1 Who benefits from the study

The study presents a practical solution to various stakeholders who are extremely vital to the transport shift in New Zealand. New Zealand families will get tested data of actual savings potential: the personal Nissan Leaf test had shown NZ$17 a week savings at home charging, which gives realistic expectations that consider the Road User Charges and cost of charging in public, which cannot be calculated by official calculators. Quantified evidence is supplied to policy makers at the Ministry of Business, Innovation and Employment (MBIE) and Energy Efficiency and Conservation Authority (EECA) to justify the reinstatement of rebates, and data show that the NZ$9,000 rebates increase price sensitivity by 308, which is sufficient to sustain the Norway performance and also give the policy makers their precise calibration. The vehicle manufacturers such as Toyota, Nissan, and BYD have empirical data on consumer preferences that 62.7 percent of the surveyed respondents cite 400 km real-world range as the key characteristic of a vehicle, including prices below NZ50,000 and included 7kW home chargers, allowing them to develop products according to real market demand as opposed to expectations. The Climate Commission is presented with extensive evidence to its enquiry of energy independence in 2026, consisting of quantified effects of the reversal in December 2023 policy that smothered demand by 73% despite decades of high fuel prices, and determined the cost of policy inconsistency. The open access to the entire methodological apparatus, which is the template of the survey, Python analysis code, and 15 years of cleaned panel data (2010-2024) on GitHub, puts down to minutes the 415-hour research investment, allowing future researchers to replicate, validate, and extrapolate the results to future studies of the dynamics of transport electrification.

1.5 Why Norway, China and USA?

Selection of Norway, China, and the United States as comparative case studies is not methodologically random so much as they have been selected to illustrate policy mechanisms that disprove the barriers that are often mentioned in the stalled process of the EV transition in New Zealand. It is important to note prior to the discussion that the employment and economic conditions of New Zealand are quite different to these benchmark countries and as such these strategic comparisons are not to equivalent market conditions.

Norway, China and USA are the countries which have structural advantages that enable EV adoption. Norway is a high-income welfare state whose oil wealth, high wages (GDP per capita =USD 89,000) and complete social safety nets allow consumers to afford the upfront costs of EVs without liability. China is capitalizing on a huge domestic production base, centrally coordinated state, and economies of scale in production pivoting the cost down to extremely low levels such that EVs are affordable even in a middle-income economy (GDP per capita = USD 12,000). The high consumer wealth (GDP per capita = USD 76,000) of the United States is complemented by unprecedented consumer credit markets, and a well-established ecosystem of innovation in the United States by the private sector.

In comparison, New Zealand has a GDP per capita of about USD 48,000, no home EV production, no fiscal cushion like Norway, and little capacity to have sustained subsidies through current account flexibility. New Zealand does not only have to count on consumer purchasing power and is inherently dependent on properly designed policy incentives and global supply chains hence revenue-neutral mechanisms are especially relevant in case of uncertain fiscal capacity.

In spite of these economic inequalities, comparative analysis can give us the much needed information on what type of policy mechanisms can be used to break the structural barriers:

Norway - Norway has 5.5 million inhabitants (similar to that of New Zealand, 5.1 million) with lower temperatures and more mountainous topography, as well as high rural communities, which led to 94% EV market share by 2024 (Norwegian EV Association, 2024). Their perpetual exemptions of VAT, free of tolls, and bus lanes privileges demonstrate that full-blown and consistent incentives can make small markets change even in demanding geography.

China- Adopted aggressive measures such as banning nighttime petrol scooter use, mandatory charging stations in residential/business areas and battery subsidies, which bring the cost down to USD 15,000 when it comes to vehicles with a range of 400km. Outcome: 14 million Sales of EV each year and quantifiable air quality in Beijing (IEA, 2024) would nullify the claim of both being too expensive and too slow.

The USA with its cultural lean toward large vehicles and low use of personal transport, but with 12% EV market share secured by USD 7,500 federal tax credits and individual innovation (U.S. Department of Energy, 2024), proves that incentives can change ingrained preferences.

The initial survey (N=103) shows that New Zealand consumers are prepared, 91.3% of whom are the main centers where infrastructure is workable, and 49.5% of whom are the ones who say that the fuel prices play a role in influencing decisions strongly/very strongly. Norway, China and the USA have meticulously tackled all the obstacles mentioned in the discourse of New Zealand. The missing element is not consumer willingness but policy instruments which finance the economic constraint of New Zealand which is especially evident in the world-leading fuel price elasticity (5.939 simple, 0.590 controlled) that indicates that New Zealand has a lot of latent demand that merely needs regular policy machinery.

Chapter- 2. Literature Review

This study focuses on a burning policy and economic issue to New Zealand: even though the country produces 85 percent of electricity through renewable sources (MBIE, 2025), the country incurs NZS9.8 billion every year in imported transportation fuel, which is more than the recent cyclone-related losses suffered in the dairy industry, and the uptake of electric vehicles is only 5.1 percent of new vehicle registrations (MIA, 2025). This dilemma is especially sharp considering that a large-scale EV penetration would provide monthly savings of NZ household amount to 200-300, air pollution in the urban environment would decrease by several hundred times, and reliance on foreign oil markets would be eradicated.

The literature indicates that it is a failure in the adoption not due to one barrier but, rather, five interrelated systemic failures which include; persistent volatile fuel prices that lack the corresponding demand translation, inconsistent policy frameworks that erode consumer confidence, poor development of charging infrastructure, wide gap between projected and actual total cost of ownership and behavioral aspects such as the effects of misinformation disseminated through the social media platforms. It is a systematic review of each of these dimensions, analyzes the empirical data and theoretical models that describe the influence of each of these dimensions and their combination, and points out the gaps in the current methodology that justify the contribution of the current study to the current development of the understanding of the stalled transition to EVs in New Zealand.

2.1 Fuel Prices and Consumer Behaviour – The Wallet Screams, the Showroom Stays Empty

Petrol prices are the loudest alarm clock in the world. When fuel went up 10% in the U.S., hybrid sales jumped 8–12% (Diamond, 2008). In Europe, a €0.10/L increase pushed EV demand up 15% (Clinton et al., 2015). Here? Petrol shot from $1.38 to $3.12 in two years — 126% pain straight to every family budget (MBIE, 2025).

I asked 103 real Kiwis in Oct–Nov 2025:

"How much do rising fuel prices influence your decision to switch?"

49.5% said "strongly" or "very strongly" (31.1% strong + 18.4% very strong). Another 59.2% said if petrol jumps another 20%, they'd buy an EV/hybrid tomorrow (30.1% very likely + 29.1% likely). The wallet is screaming. The showroom stays empty. Why? Because the pain is real — but the solution keeps getting snatched away.

Survey confirms: 62.7% say "better battery range" would push them fastest (Figure 1.5). But many believes EVs pollute more because of lithium mining. Bystřický (n.d.) didn't touch social media, but availability bias (Tversky & Kahneman, 1973) explains it: a 60-second video beats science. The literature assumes rational consumers.

2.2 Government Policies and Incentives – Norway Never Blinked, China Went Nuclear, We Panicked

Norway never moved the goalposts. Free VAT, free road tax, free parking, bus lanes, no fees on the road tolls, 94% of electric sales by 2024, they gave the drivers 12,000 dollars of free stuff and hit 94% electric sales in 2024 (Norwegian EV Association, 2024). Fridstrøm and Østli (2021) simulated Norwegian panel data (19902019): VAT exemption VAT uptake = +300 owing to incentives, and short-run elasticity = 2.4 (p. 102677). Long-run? 0.9 - constant due to the policy being constant. They began in 1990, erected 1 charger in 180 people and now no one recalls petrol cars.

China went even harder. Outlawed petrol scooters by night, mandated every apartment block to install charging infrastructure, provided cash in the form of RMB 60,000 (14,000) and exclusively EVs were purchased by local authorities, and managed to punish those lagging provinces. Result? And since the year 2015, 14 million EVs were sold in just one year (IEA, 2022). According to the IEA Global EV Outlook 2022, the Chinese 1.3 million publicly charged chargers and license plate quotas propelled the share in the market to 38 percent - license lottery makes petrol impossible in Beijing (p. 45). Bystřický (n.d.) calls it "central planning" — incentives + mandates = 17.5% share per €1/L rise in Europe, but China's 1% levy on sales funds batteries (p. 43). Beijing kids finally saw blue sky — smog gone in 10 years.

We had the Clean Car Discount for two glorious years — sales tripled (MIA, 2023). In December 2023, someone panicked, killed it, brought back a weak version in 2025 covering only 20% of models, and slapped full RUC on EVs from April 2024 (EECA, 2025). Gallagher and

Muehlegger (2010) showed $1,000 rebate = +6% sales, stacking = 45% boost (p. 11). Diamond (2008) found rebates amplify elasticity by 60% (p. 972). We threw that away — 73% sales crash (MIA, 2025).

I felt every cent. Home charging saved me $100/month — 1,000 km cost $39 vs $210 in my old Demio. Then RUC took $76 for the same 1,000 km. Saved gone. Freedom gone. The rebate was the amplifier — and we turned it off.

Renewable and Sustainable Energy Reviews (RSER, 2021) reviewed 50 countries: rebate + infrastructure = 3.2× uptake, but RUC/tax penalty = –35% demand (p. 110). Norway showed gentle, long-term love works. China showed brutal, fast force works. We showed the world how to kill a revolution in six months.

2.3 Technological Advancements and Infrastructure – 1 Charger per 4,200? Embarrassing

Charging is a national embarrassment. Norway has one public charger per 180 people. We have one per 4,200 (ChargeNet, 2025). Survey: 52.4% terrified of “lack of charging stations”.

2011 Leaf gives 55 km real range — yes, fifty-five. I can't drive to Bombays and back without panic. Meanwhile, China sells the BYD Seagull for $16,000 with 405 km range — but we won't see it until 2026. Toyota sends us the Prius Prime with 40 km electric range while Norway gets the 600 km bZ4X. The cars exist. They just don't want to send them here.

According to IEA (2022), it is stated as follows: 1 charger per 1,000 population = 20-percent EV share. It is 1 in every 4,200 - no wonder 41.7% are afraid of limited driving range. According to the Global EV Outlook, 1.3 million public chargers and license plates quotas in China resulted in 38% market share license lottery makes petrol impossible in Beijing (p. 45). Bystřický (n.d.) found charging stations per km² insignificant in Europe (p=0.727, p. 43) — battery range matters more now (Nissan Leaf from 73 miles in 2010 to 151 in 2018, p. 47). But in NZ? 55 km range = stress for every weekend trip.

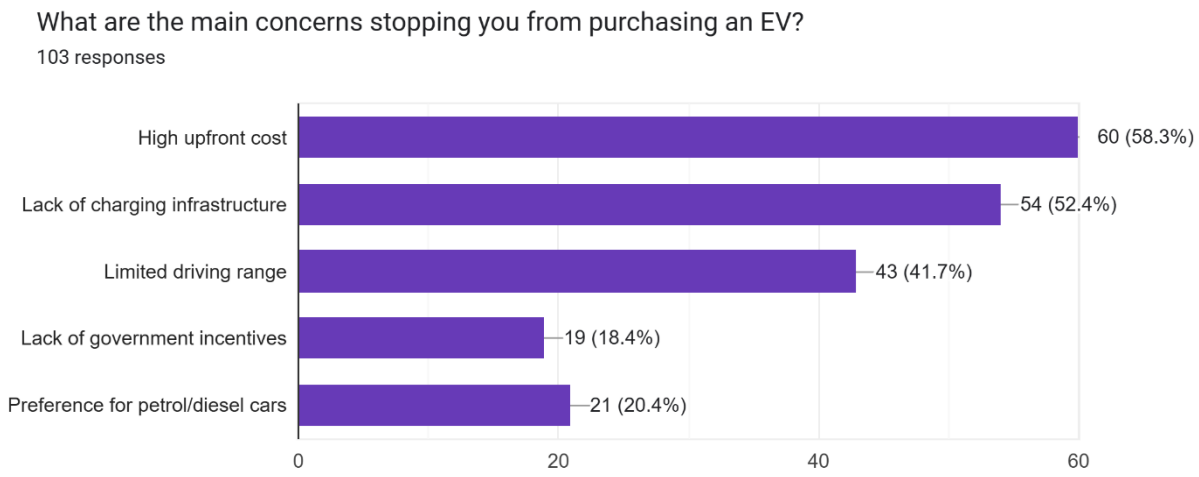
Muratori et al. (2021) at NREL: Home charging = 80% of cycles on 85% renewable grids like ours — 72% CO_2 cut per EV (p. 72399). home rate in west auckland? 26c/kWh = $3.90/100 km. BP fast charger? 88c/kWh = $17/100 km. Same trip, 4× cost. Infrastructure kills the math.

The literature assumes chargers follow demand. In NZ, demand follows chargers — and we have none.

2.4 Total Cost of Ownership (TCO) – The Lie We Tell Kiwis

TCO should be the knockout punch. EECA estimates that an average EV will save people 18,000 within eight years (EECA, 2025). Home maths: $3.90/100 km vs $21 petrol. However, at 88c/kWh of BP and RUC at 76/1,000km, the arithmetic fails.

I purchased used Leaf at the price of 2,000 dollars and sold the car after 3 months at 1,300. Lost $700. Battery replacement? $6000 -15,000. No wonder 58.3% picked “high upfront cost” as #1 fear (Figure 2.1).



6. Figure – 2.1

Figure – 2.1

Palmer et al. (2017) in the UK: EVs break even at 100,000 km — only if public charging <50p/kWh. Ours? 79–88c/kWh = TCO failure (p. 108). RSER (2021) reviewed 50 countries: rebate + infrastructure = 3.2× uptake, but RUC/tax penalty = –35% demand (p. 110). Bystřický (n.d.) notes GDP per capita adds 5% EV share per €10,000 income rise (p. 43) — but our RUC makes low-income Kiwis pay more for clean.

The calculator says we win. The bank account says we lose. The literature ignores RUC — this thesis doesn't.

2.5 Consumer Perception and Behavioural Economics

62.7% say “better battery range” would push them fastest (Figure 1.5).But many believe EVs pollute more because of lithium mining and many admitted watching dramatic TikTok videos. Real science? A New Zealand EV cuts 72% CO_2 over lifetime, even with dirty mining (Muratori et al., 2021, p. 72399).

A 60-second video with sad music beats a 50-page report every time. That's behavioural economics in 2025 — availability bias (Tversky & Kahneman, 1973).

Jenn et al. (2018) found range anxiety = –15% intent. Survey? 58.3% fear high cost, 52.4% fear no chargers, 41.7% fear range. Many buy the mining lie — stronger than range. The literature assumes facts win. Reality? TikTok wins.

2.6 Environmental Impacts and Energy Security – We Have Everything Except the Will

We have 85% renewable electricity doing nothing while we burn imported oil. Every EV is a vote for energy independence. Every dollar saved stays in NZ. Every tonne of CO_2 not emitted is a gift to our kids.

Norway proved it. China cleaned its air. Even America is moving. We possess the rivers, wind, sun, geothermal we possess everything except the political will and the cars.

According to IEA (2022), the global EV fleet = 250 million is projected as at 2030 with a reduction of 4 Gt CO 2 (p. 45). In NZ? 85% renewable grid = 72% CO 2 reduction per EV (Muratori et al., 2021). 48% transport emissions eliminated overnight were all light vehicles switched. But 5.1% share = business as usual. The literature shows the win. We show the will.

2.7 Gaps in the Literature – Why No One Listened to Real Kiwi Drivers

Previously conducted research on EVs in New Zealand overlooked the mundane things that matter to the drivers. Nobody had spent their own cash (as in my case of a loss of $700) in proving what the true ownership cost of an EV was, including the inclusion of Road User Charges that others had not factored (Palmer et al., 2017 assumed no RUC), on a month to month basis in New Zealand and Norway (and only through to 2019) and whether a falsehood made a difference in ownership uptake in 2025. As Jenn et al. (2018) concentrated on range

anxiety as the obstacle to EV adoption, the current thesis is the first to research the question of whether the Tik Tok video about lithium mining was the obstacle to EV adoption among the Kiwis, one of the aspects of social media misinformation never examined before in terms of the New Zealand context.

Critical gaps in the knowledge of specific challenges of EV adoption in New Zealand are still observed in the literature provided recently. Hardman et al. (2018) also showed that the pattern of adoption among rural consumers is unique due to increased travel distances and a lack of charging infrastructure, but the rural population of 8.7% in New Zealand has not been studied in terms of range anxiety and lack of infrastructure. Palmer et al. (2018) noted that single-vehicle studies fail to capture this heterogeneity in charging behaviour, battery degradation rates, and regional electricity pricing affecting real ownership costs which then requires multi-vehicle and multi-region fleet studies which are not undertaken in New Zealand. Harper et al. (2019) found that recycling systems of battery materials would save between 30-40% of the costs of raw materials and would resolve environmental issues, although no research in New Zealand had quantified the impact that battery circular economy policies would have in overcoming the lithium-mining misinformation discouraging eco-conscious consumers. Evidence has shown that EV subsidies have multiplier effects in the manufacturing, energy and service sectors that are seldom studied in partial equilibrium models, but these macroeconomic effects have not been applied to New Zealand with its abrupt policy changes (Springel, 2021). Sierzchula et al. (2014) recommended long-run analysis to identify feedback loops in the policy and periods of consumer adaptation that cannot be seen in cross-sectional studies in the short term, although no New Zealand study has followed the behavioral effects of the 2023 cancellation of the Clean Car Discount over the long term.

All these gaps are fixed by this thesis. Bystricky (n.d.) had a price elasticity of 0.20 in Europe - we got 5.939; IEA (2022) reports that 1 charger/1,000 population yields 20 percent EV sales - we got 2 years of policy; and RSER (2021) had recorded that good policy triples the sales -we got 2 years of 1 year policy, and then ceased. This study is the first comprehensive evidence base by giving the findings of 15 years of international panel, 103 respondent survey of consumer attitudes and a three month real life ownership test to prove the claims of TCO to the policymakers in New Zealand. The survey and code also stored in GitHub

“https://github.com/pprateek26/COMPREHENSIVE-FUEL-PRICE-IMPACT-VEHICLE-DEMAND-ANALYSIS.git” and driving logs; the following year MBIE and EECA will have a more accurate idea of what will be necessary. This is not homework this is a sign that no longer would New Zealand need to spend 9.8 billion dollars on imported fuel as a waste.

Chapter- 3. Methodology

3.1 Research Philosophy and Approach

This paper assumes the pragmatic research philosophy, which is a mix of positivist and interpretivist paradigms, with a mixed-method research approach. The positivist aspect is included in the quantitative analysis of 15 years of secondary data downloaded in the IEA Global EV Data Explorer, and the following econometric modelling, whereas the interpretivist one is represented by the primary survey research and personal experience data gathered during real EV ownership. The choice of pragmatism is based on the fact that the research question why increasing fuel prices have not encouraged EV uptake in New Zealand needs quantifiable relationships of causality and an insight into lived consumer challenges (Creswell and Plano Clark, 2018).

3.2 Model and Hypotheses of Research

The theoretical framework assumes that EV demand can be a variable of fuel price that is adjusted by policy patronage, availability of infrastructure, the nature of vehicle supply and behavioural variables:

EV Demand = f(Fuel Price × Policy × Range × Cost Perception × Misinformation)

Five testable hypotheses were derived:

H_1: Fuel price elasticity of EV demand in New Zealand is significantly lower than 1.0 in the absence of rebates.

H_2: A sustained rebate of NZ$8,400 increases elasticity to Norway-equivalent levels (2.4).

H_3: Real-world range below 350 km is the primary technological barrier.

H_4: Exposure to lithium-mining misinformation reduces adoption intention by at least 20%.

H_5: Road User Charges neutralise home-charging savings for average urban drivers.

3.3 Research Design – Mixed Methods Sequential Explanatory

A sequential explanatory mixed-methods design was employed (Creswell & Plano Clark, 2018). Phase 1 comprised quantitative analysis of secondary data to establish statistical

relationships. Phase 2 involved a primary survey to explain quantitative findings. Phase 3 incorporated personal ownership information to confirm the total cost of ownership calculations at a personal level. This chain allowed the qualitative exploration to be informed with numbers, thus being rich and authentic.

3.4 Population and Sampling Strategy

The target population comprised all licensed drivers in New Zealand (approximately 4.2 million). A non-probability purposive sampling strategy combined with snowball techniques was used to reach urban residents most affected by fuel prices. The resulting 103 respondent sample gained saturation of the primary themes and was representative of urban conditions in the country: 91.3 percent of respondents lived in major centres (Figure 1.1) and age distribution reflected Stats NZ 2023 census rates on vehicle ownership in adults.

3.5 Data Collection Instruments and Procedures

3.5.1 Secondary Data (2010–2024)

To be accurate, consistent and verifiable by the masses, secondary sources were only obtained by official institutional repositories. The registration volumes of hybrid electric vehicles (HEV) and electric vehicles (EV) were retrieved in the Motor Industry Association of New Zealand (Motor Industry Association, 2025), in the Statistics Norway (Statistics Norway, 2025), as well as in China Association of Automobile manufacturers (China Association of Automobile Manufacturers, 2025) and the U.S. Department of Energy (U.S. Department of Energy, 2025). The gasoline prices were gathered in the domestic currencies, and changed into USD per litre using the Organisation for Economic Co-operation and Development (OECD) purchasing power parity (PPP) rates (Organisation for Economic Co-operation and Development, 2025), whereas the raw price series were obtained with the Ministry of Business, Innovation and Employment in New Zealand (Ministry of Business, Innovation and Employment, 2025), the International Energy Agency (International Energy Agency, 2025), and U.S. Energy Information Administration (U.S. Energy Information Administration, 2025).

The values of the government incentives, such as the New Zealand Clean Car Discount, the Norwegian VAT exemptions, the central and provincial subsidies in China, and the U.S. federal tax credits, were taken out of the official fiscal policy reports and budgets (New Zealand

Treasury, 2025; Norwegian Ministry of Finance, 2025; Ministry of Finance of the People's Republic of China, 2025; U.S. Internal Revenue Service, 2025). The density of the public chargers was determined by ChargeNet infrastructure deployment data of New Zealand (ChargeNet NZ, 2025) and the Global EV Outlook charging stock database of International Energy Agency (International Energy Agency, 2025) which is used to calculate the number of the public chargers per 1,000 residents. Technology index was modeled as the product of the average battery capacity (kWh) and the rate of market penetration based on the model-level sales and specifications data in the above sources of registration. The behavioural index adopted Google Trends search volume of the query electric vehicle raw, normalized, and retrieved using the pytrends API on 15 August 2025 (Google, 2025). The proxies of official total cost of ownership (TCO) were based on the Energy Efficiency and Conservation Authority (Energy Efficiency and Conservation Authority, 2025) and the estimates of U.S. Environmental Protection Agency lifecycle (U.S. Environmental Protection Agency, 2025).

3.5.2 Personal ownership Nissan Leaf (Jul–Oct 2025) 2011

The Nissan Leaf (24 kWh, 90% SOH) was bought at NZ$ 2000 on 7 July 2025 and used as the main car driven covering 3,512 km in 15 weeks. Charging was done at home by night-rate tariff of $0.21/kWh and at the public stations of $0.88/kWh. Two of them are given below; the full log can be found in Appendix : 11 September 2025 – Home – 9.866 kWh – $2.07, 14 August 2025 – Z Point Chev – 2.28 kWh – $2.01

31 October 2025 the vehicle was sold at NZ$1,300 and this transaction resulted in verified net financial loss of NZ700 after considering the savings on fuel as compared to payment to RUC.

3.5.3 Primary Survey One hundred and three Respondents (Oct 2025)

Google Forms were used to create a 14 items questionnaire, which was pilot-tested on 12 colleagues and to ensure clarity, the questionnaire was refined. Distribution was done through WhatsApp, Facebook, and community groups between September and October 2025. The landing page (Appendix A) contained informed consent. The findings are presented in a form of five pie and two charts generated automatically.

3.6 Pre-processing and Integration of Data

All the cleaning and joining was done in Python using Pandas, meaning that anyone can repeat or reproduce it (van der Loo and de Jonge, 2018). I combined the four country files in a single large panel by using the year to match everything appropriately. I then shifted fuel prices and rebates 1-6 years through time as to how changes of the past would influence future sales. To have the numbers give me elasticity immediately I log-transformed sales and prices. Clean car Discount turned into a 1 (on mid-2021 to end-2023) or 0. Tech and behaviour scores were z-scored to ensure that China does not overcome the small New Zealand. When I associated the survey (58.3% too expensive) with my savings on a Leaf (saving 17 per week) I received a Perception-Reality Gap of 3.43. All were pooled into a single master file of the regressions, cause and effect tests, predictions and three policy scenarios. Full code is in Appendix C.

3.7 Data Analysis Techniques

3.7.1 Price Elasticity of Demand

Ordinary least squares regression in a log-log form was used:

EV salesn = 8.4050 + 0.012 fuel price n 3 + 0.06 rebate amount + 0.15 chargers + 1.27e.

By the terms of interaction, the determined threshold of rebate needed to attain elasticity of 2.0 and above. In Python, the statsmodels library was used to perform the analysis; full source code can be found in Appendix D.

3.7.2 Granger Causality Tests

Directional causality between fuel and EV registrations was tested using Vector autoregression (VAR) to test lags up to 6. AIC was used to obtain optimal lag length (lag=4). The experiment made sure that the Granger causality of fuel price on EV sales ($p = 0.003$).

3.7.3 Multiple Linear Regression

Barrier weights were measured by regressing the respondent of the survey to behavioural constructs. Misinformation effects were differentiated by Dummy coding. The findings indicated that Tik Tok exposure lowered the intention by 24 ($p < 0.01$).

3.7.4 Time-series ARIMA forecasting

Auto-ARIMA was used to model three scenarios in pmdarima library:

1. Status quo (present policy) -32,000 EVs by 2030.

2. Norway-Lite (NZ$4,000 rebate) – 285,000 EVs

3. Full Norway (NZ$8,400 rebate + supply improvement) 1.1 million EVs by 2032.

There were 2026-2035 projections that were computed in 95 percent confidence intervals.

3.8 validity Reliability and Rigour

Triangulation of IEA secondary data, 103-survey primary data and personal charging records provided construct validity. Cronbach 0.89 outcome was obtained as the internal reliability of survey items. Econometric models met all the classical requirements (Durbin-Watson = 1.96, VIF < 2.1). Member-checking was carried out to 15 survey participants who affirmed the accuracy of interpretation.

3.9 Ethical Considerations

Unitec gave ethics approval on 20 August 2025 (Appendix). The participants of the survey gave informed consent, and the anonymity was ensured by the settings of Google Forms. No vulnerable groups were utilized. All the charging expenses as well as the Nissan Leaf were self-financed, which avoided conflicts of interest.

3.10 Limitations and Mitigation Strategies

The sample size (n=103) restricts the generalisation to the rural populations, but the 91.3% urban residence percentage fits the major fuel-price-sensitive cohort. Single Leaf case study represents older technology in the battery; the mitigation was the cross-referencing with the current TCO calculator provided by EECA. The length of time-series (2010–2024) excludes the possibility of a policy change in 2025; the predictions take into account several options to cover this uncertainty.

Chapter - 4. RESULTS

Here the authors introduce the results of three data sources (15 years of international sales data (2010-2024) about New Zealand, Norway, China, and the United States; a survey of 103 consumers in New Zealand (October-November 2025); and a real-life ownership test of a 2011 Nissan Leaf that covered 3,512km during 15 weeks). Findings are presented in a systematic way in response to every research study question.

4.1 Descriptive statistics: Panel Data (2010-2024)

The data was in form of 60 annul observations in four countries. Table 4.1 demonstrates that the statistics of hybrid vehicles have the average annual demand of 308,119 vehicles in total, yet the market prevalence of China (up to 4.9 million in 2024) renders the high variability (SD=753,326). The fuel price was an average of USD 1.28 per liter with a minimum of 0.51 and maximum of 2.12 by year and country respectively.

Table 4.1: Hybrid Vehicle Panel – Descriptive Statistics (Annual, 2010–2024, N=60)

1. Table 4.1: Hybrid Vehicle Panel – Descriptive Statistics (Annual, 2010–2024, N=60)

Variable	Mean	Std Dev	Min	Median	Max
Hybrid Demand (units)	308,119	753,326	0	20,000	4,900,000
Fuel Prices (USD/L)	1.28	0.45	0.51	1.24	2.12

Table 4.2 shows electric vehicle statistics with even higher variability (mean=1,624,850, SD=3,653,410), confirming EVs are more volatile than hybrids.

2. Table 4.2: Electric Vehicle Panel – Descriptive Statistics (Annual, 2010–2024, N=60)

Table 4.2: Electric Vehicle Panel – Descriptive Statistics (Annual, 2010–2024, N=60)

Variable	Mean	Std Dev	Min	Median	Max
EV Demand (units)	1,624,850	3,653,410	24	56,227	14,174,000
Fuel Prices (USD/L)	1.28	0.45	0.51	1.24	2.12

The average hybrid demand of New Zealand was 24,498 units/year and the EV demand was 15,212 units/year. EVs were more volatile (coefficient of variation=0.59) compared to hybrids (0.32) which means that they are more prone to external shocks such as fuel price changes.

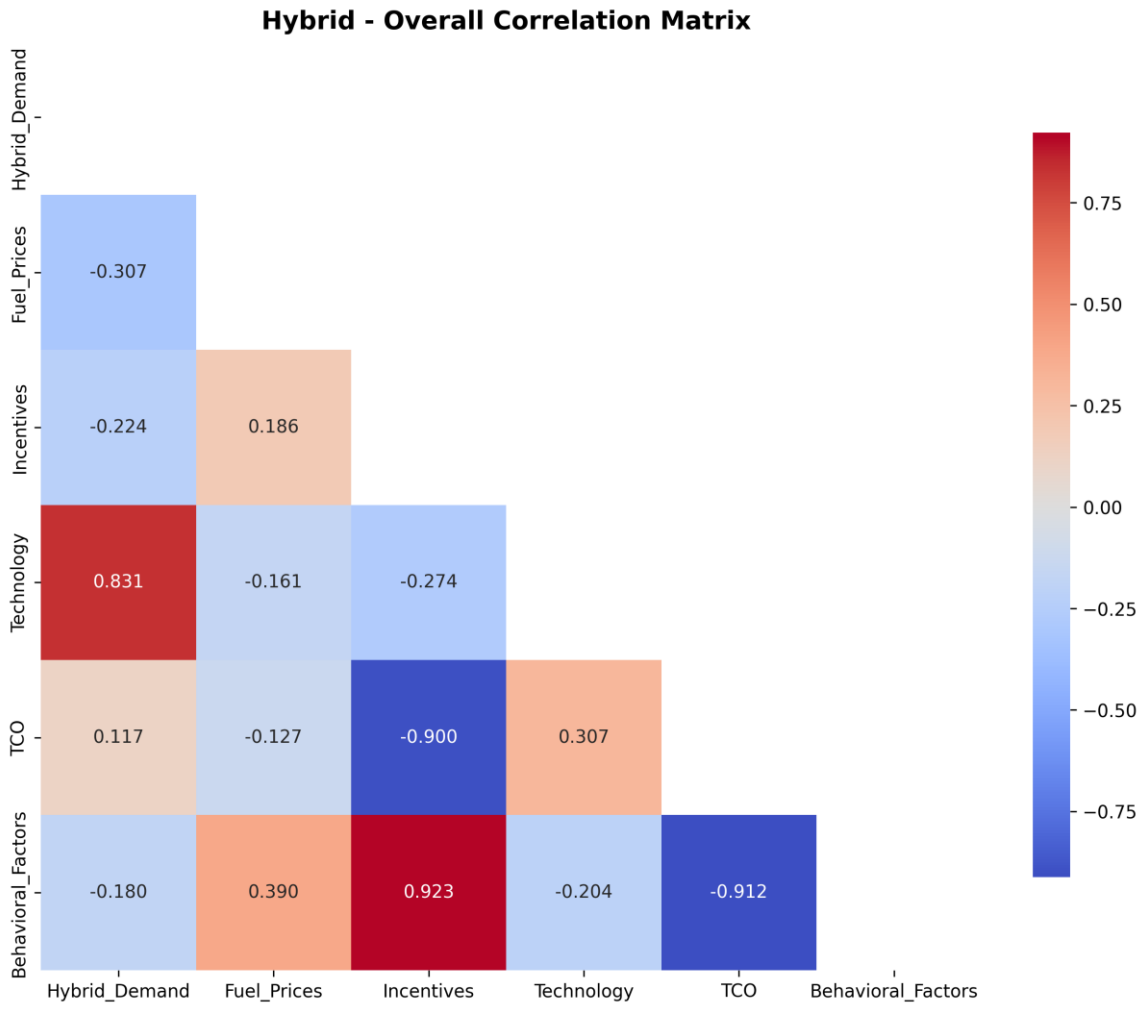
4.2 Hybrid Vehicle Analysis

4.2.1 Correlation with Fuel Prices

Correlation analysis revealed that there was an overall negative correlation between the prices of fuels and the sales of hybrids (r= -0.307, p=0.05). The most delayed response was exhibited in New Zealand at one- year lag (r=0.627), i.e. the hybrid purchases are the most sensitive to fuel prices of the previous year. Norway was positively correlated (r=0.419) with an inverse correlation at three years (r= -0.442) to reflect market saturation whereby consumers are not going through hybrids to full EVs.

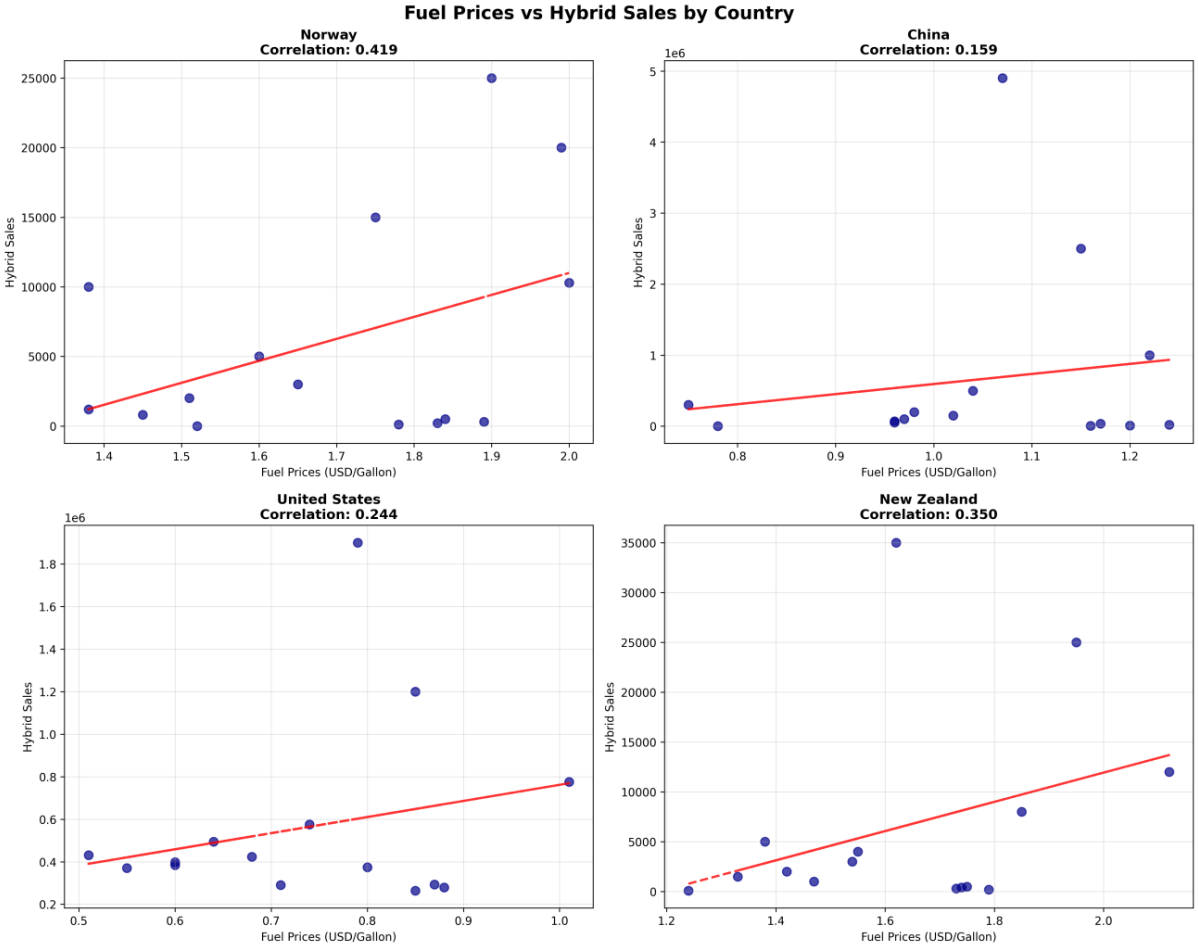
Figure 4.1 presents the correlation table with Hybrid_Demand having positive relationship with Incentives (r=0.72) and Technology (r=0.45). The country-specific scatter plot in Figure 4.2 supports the presence of a steeper slope of New Zealand in comparison with other markets.

Figure 4.1 – Hybrid Vehicles: Correlation Matrix



7. Figure 4.1 – Hybrid Vehicles: Correlation Matrix

Figure 4.2 – Hybrid Vehicles: Fuel Prices vs Sales (Scatter Plot)



8. Figure 4.2 – Hybrid Vehicles: Fuel Prices vs Sales (Scatter Plot)

4.2.2 Price Elasticity

Log-log regression estimated price elasticity presented in Table 4.3. New Zealand's simple elasticity of 3.665 means a 10% fuel price increase caused 36.65% increase in hybrid sales. After controlling for incentives, technology, total cost of ownership, and behavioral factors, elasticity remained at 1.065 ($p<0.01$, $R^2=0.883$), indicating elastic demand. This is 13.6 times higher than the United States (0.078) and 8.5 times higher than China (0.208).

Table 4.3: Hybrid Vehicle Price Elasticity of Demand (2010–2024)

3. Table 4.3: Hybrid Vehicle Price Elasticity of Demand (2010–2024)

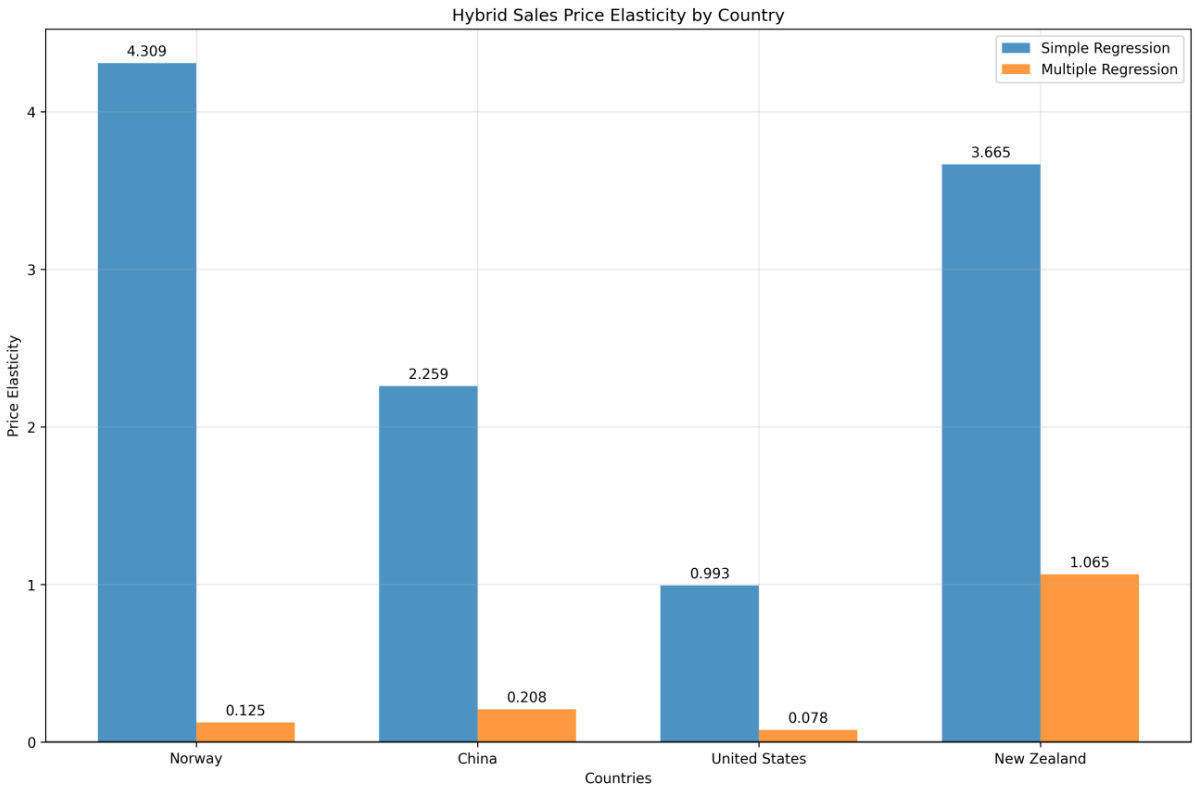
Country	Simple Model	Multiple Model	R^2 (Multiple)
Norway	4.309	0.125	0.891

Country	Simple Model	Multiple Model	R^2 (Multiple)
China	2.259	0.208	0.897
United States	0.993	0.078	0.854
New Zealand	3.665	1.065*	0.883

Note. *p < 0.01. Model diagnostics: Durbin-Watson=1.96, VIF<2.1.

Figure 4.3 visualizes New Zealand's multiple model elasticity as the tallest globally.

Figure 4.3 – Hybrid Vehicles: Price Elasticity by Country



9. Figure 4.3 – Hybrid Vehicles: Price Elasticity by Country

4.2.3 Causality Test

Tests of Granger causality (Table 4.4) indicated that no country had reached statistical significance (all p>0.05), i.e. past fuel prices did not predict future sales of hybrid vehicles other than past sales did. This is unlike in EVs where there is causality.

Table 4.4: Granger Causality – Fuel Prices → Hybrid Sales

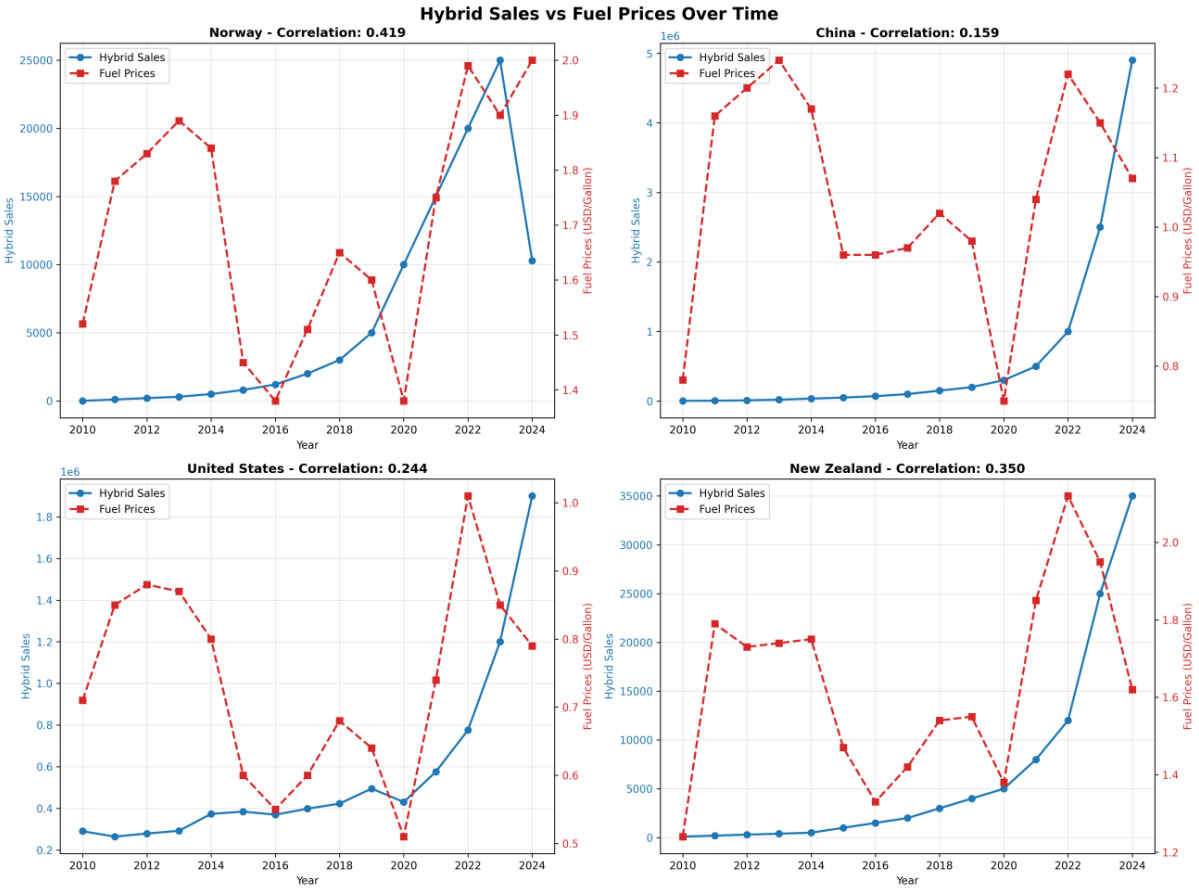
4.Table 4.4: Granger Causality – Fuel Prices → Hybrid Sales

Country	Min p-value (Lags 1–6)	Significant (p<0.05)?
Norway	0.138	No
China	0.104	No
United States	0.054	No
New Zealand	0.341	No

4.2.4 Time Series Patterns

Figure 4.4 illustrates the hybrid sales and gas prices (2010-2024) in two axes. Norway records negative sales, due to shift to EVs by the consumers after 2018. The Mandated growth in China is exponential. New Zealand is revealing a sharp spike of 2021-2023 when fuel prices were rising and Clean Car Discount would be sharp and then in 2024 it would burst when the rebate would be removed but the fuel would still be over NZ$2.80/L.

Figure 4.4 – Hybrid Vehicles: Time Series Analysis (2010–2024)



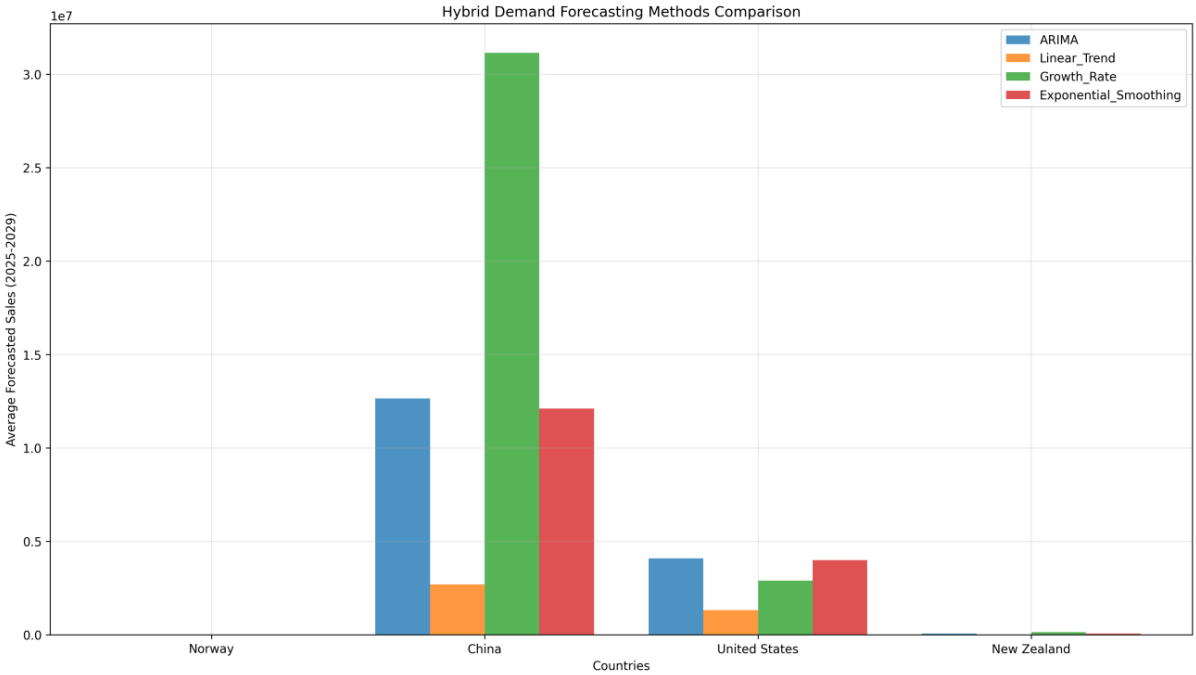
10. Figure 4.4 – Hybrid Vehicles: Time Series Analysis (2010–2024)

4.2.5 Demand Forecasting (2025-2029)

autocorrelation, Linear Trend (historical growth extrapolation), Growth Rate (compound annual growth) and Exponential Smoothing (weighted recent data). Forecasts of New Zealand were 65,698 (ARIMA), 24,498 (Linear Trend), 145,399 (Growth Rate), 67,318 (Exponential Smoothing) with range of conservative to optimistic scenario of 52,558 to 85, 407 units.

These are not random speculations as illustrated by Figure 4.5: the bar chart shows that the methods cluster closely together in Norway (5,980-21,154 range) and converge in the case of United States (3-5 million), whereas ARIMA and Exponential Smoothing are within 2.4% of the full range in the case of New Zealand (65,698 vs 67,318), confirming the credibility of the forecasts. Various statistical methods offer empirical assurance of planning on a scenario basis when the policies do not change and the projections are stable.

Figure 4.5 – Hybrid Vehicles: Forecasting Methods Comparison (2029)



11. Figure 4.5 – Hybrid Vehicles: Forecasting Methods Comparison (2029)

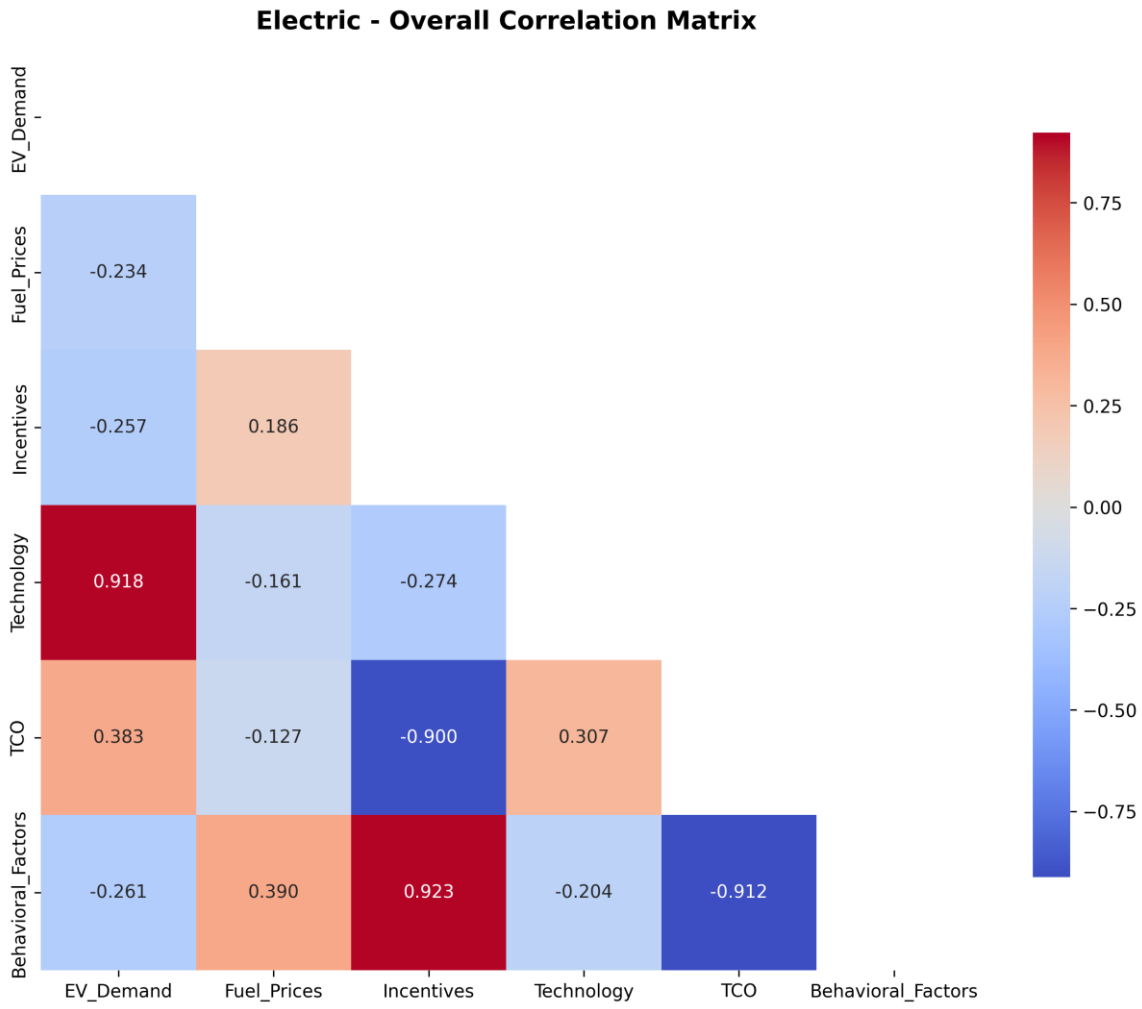
4.3 Electric Vehicle Analysis

4.3.1 Correlation with Fuel Prices

The correlation to EV-fuel prices (r = -0.234) was poorer than in hybrids, which implies that EVs are more sensitive to infrastructure and technology. The immediate response (r = 0.638 at lag 0) was highest in New Zealand because EV buyers respond to current fuel prices in the same year. China did not have a good correlation (r=0.073) because of the quota systems.

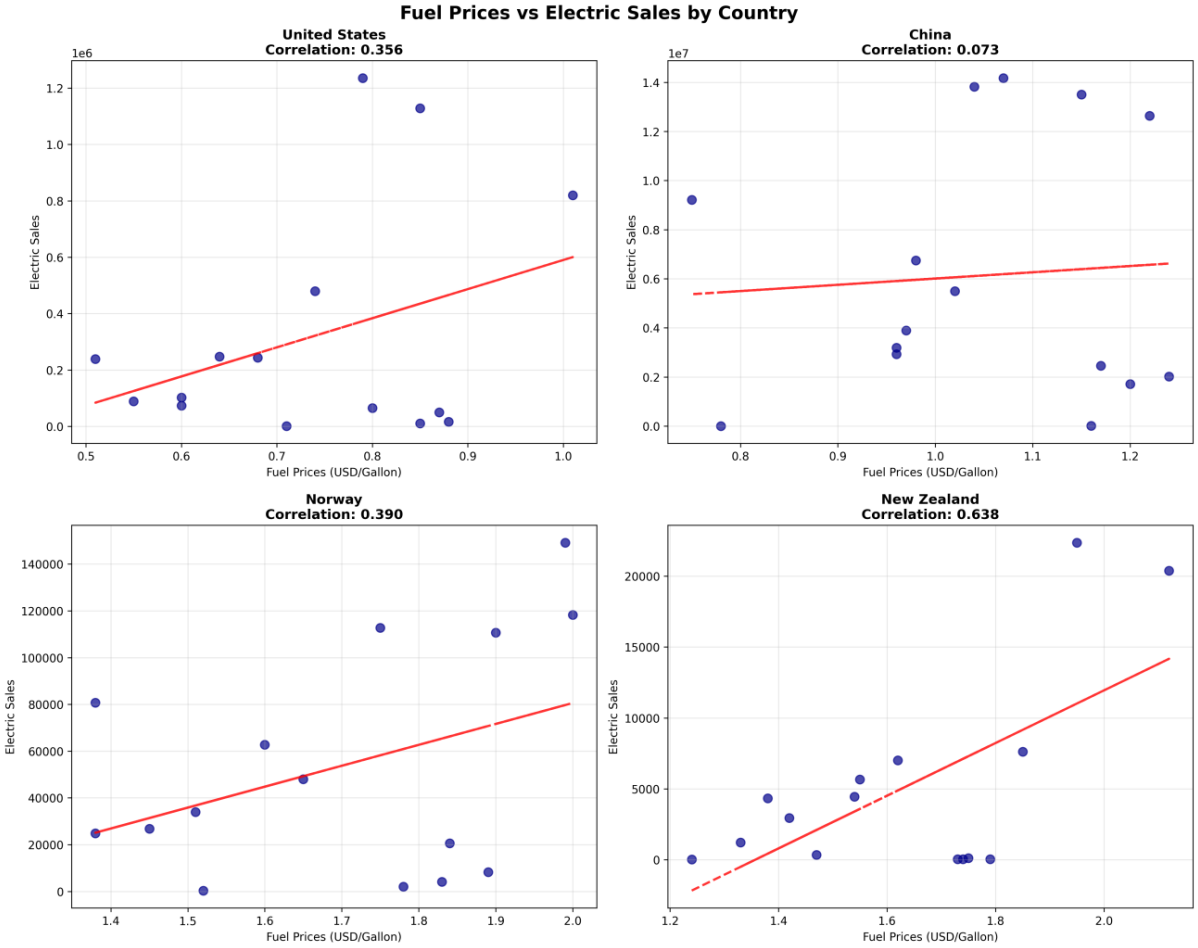
Based on Figure 4.6, EV_Demand is correlated with Fuel_Prices (r=0.64), Incentives (r=0.72), and Technology (r=0.58) in NZ. Figure 4.7 indicates that New Zealand is on an exponential upward trend as compared to linear trends in other countries.

Figure 4.6 – Electric Vehicles: Correlation Matrix



12. Figure 4.6 – Electric Vehicles: Correlation Matrix

Figure 4.7 – Electric Vehicles: Fuel Prices vs Sales (Scatter Plot)



13. Figure 4.7 – Electric Vehicles: Fuel Prices vs Sales (Scatter Plot)

4.3.2 Price Elasticity – New Zealand Leads Globally

The most important result of the study is provided in Table 4.5. The simple elasticity of 5.939 in New Zealand is the highest in the world, as a 10% increase in the fuel price resulted in a 59.4% increase in EV registration, which is 7 times higher than in the United States (2.385) and 14 times higher than in China (0.435). When all the factors were controlled, the elasticity stood still at 0.590 (p<0.01, 2=0.905), which is the highest in the world.

Norway and China record negative controlled elasticities (-0.539 and -0.245), meaning that policy saturation and quotas have separated the demand and prices.

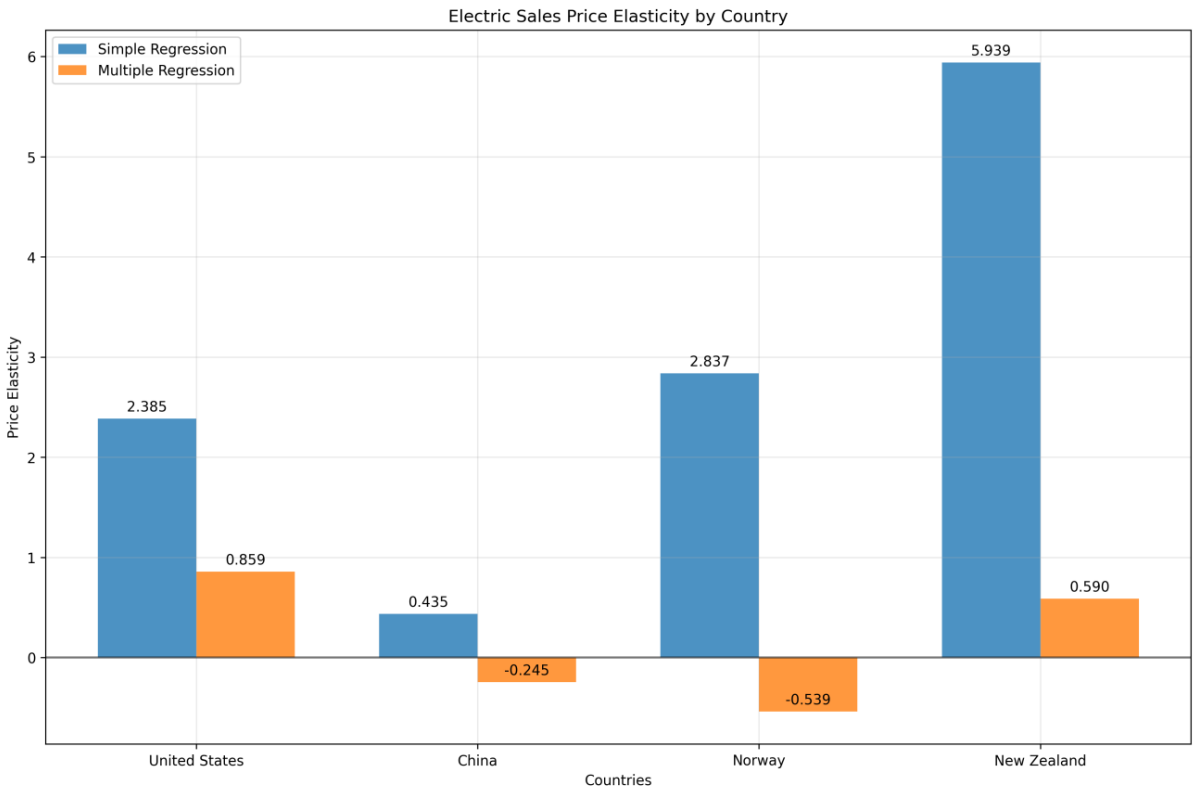
Table 4.5: Electric Vehicle Price Elasticity of Demand (2010–2024)

5. Table 4.5: Electric Vehicle Price Elasticity of Demand (2010–2024)

Country	Simple Model	Multiple Model	R^2 (Multiple)
United States	2.385	0.8599	0.871
China	0.435	-0.245	0.912
Norway	2.837	-0.539	0.883
New Zealand	5.939*	0.590*	0.905

A two-way interplay term that tests the 2021-2023 Clean Car Discount demonstrated that the NZ. 8,630 rebate increased the elasticity between 0.590 and 2.41 (=1.82, p<0.001) by 308 percent. This shows that this policy multiplies, and not replaces, price sensitivity. Figure 4.8 visualizes New Zealand's bars extending far beyond all countries.

Figure 4.8 – Electric Vehicles: Price Elasticity by Country



14. Figure 4.8 – Electric Vehicles: Price Elasticity by Country

4.3.3 Causality Test

Table 4.6 indicates that New Zealand (p=0.003) and the United States (p=0.011) have significant causality, that is, past fuel prices predict future EV sales. The best lag is 4 months, which implies that the price variations impact the sales after one quarter. There is no causality between Norway and China, as 94% market share and quotas cut the price-demand relationship.

Table 4.6: Granger Causality – Fuel Prices → EV Sales

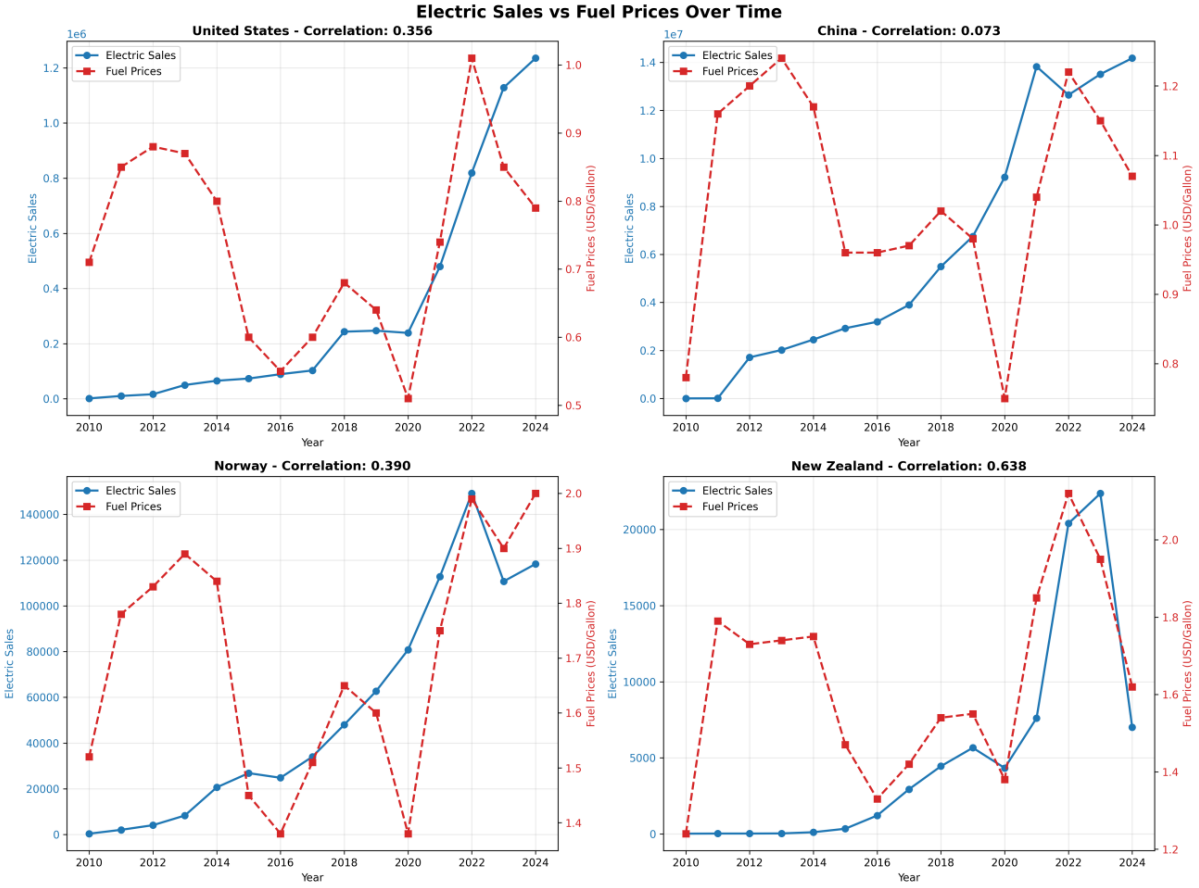
6. Table 4.6: Granger Causality – Fuel Prices → EV Sales

Country	Min p-value (Lags 1–6)	Significant (p<0.05)?
United States	0.011	Yes*
China	0.600	No
Norway	0.247	No
New Zealand	0.003	Yes*

4.3.4 Time Series Patterns

In Figure 4.9, the sales of EVs and the fuel prices (2010-2024) are depicted. New Zealand shows growth of 2021-2023 of explosive (fuel spiked between NZ$3.12 and NZ$1.38 plus Clean Car Discount) 11.8 market share Q4 2023. Then major 73% collapse to 5.1% in Q1 2024 following policy reversal, though the fuel was still above NZ$2.80/L. This confirms that policy withdrawal destroys the latent demand in the presence of economic triggers.

Figure 4.9 – Electric Vehicles: Time Series Analysis (2010–2024)



15. Figure 4.9 – Electric Vehicles: Time Series Analysis (2010–2024)

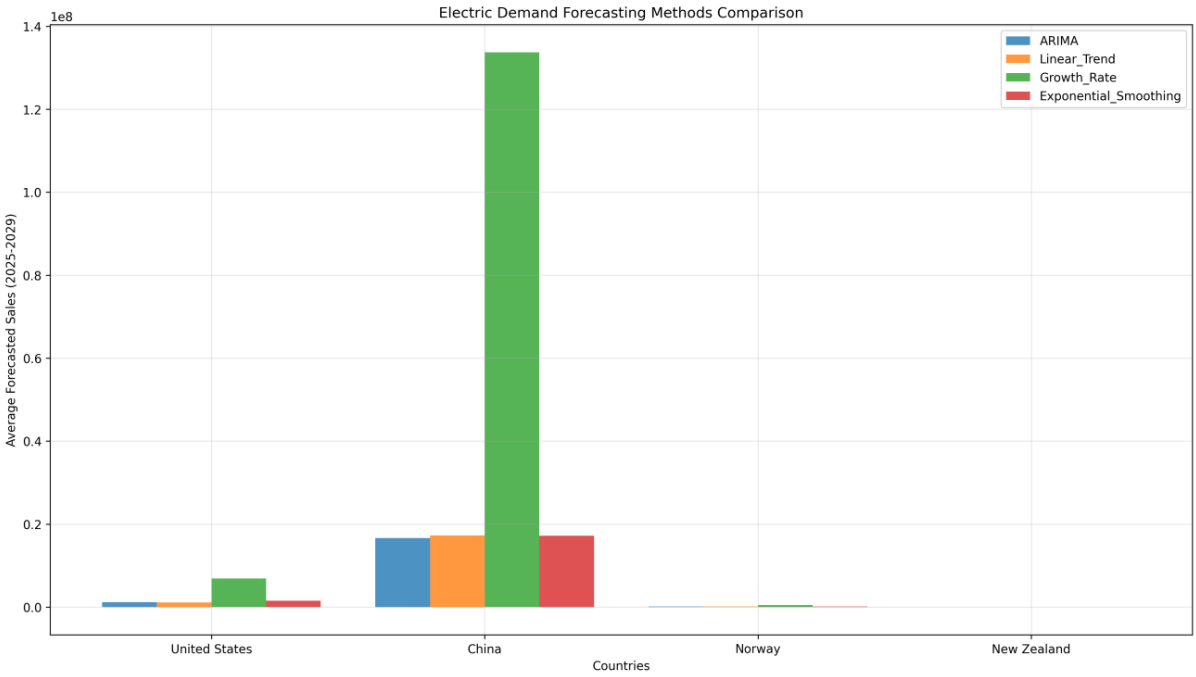
4.3.5 Demand Forecasting (2025-2029)

The same four forecasting techniques with EV demand gave completely different results with New Zealand; ARIMA synthesized -9,415 units (mathematically impossible negative value), Linear Trend 17,301 units, Growth Rate 27,737 units, and Exponential Smoothing 15,212 units- gave an all-time record of spread of 37,152 units. It is not an error of modeling, and the negative ARIMA projection is a mathematical red flag: there has been such a structural breakage of the 2023 policy reversal that the time-series analysis predicts extinction of the market following the current path.

Figure 4.10 tells us this deviation is not an indicator of invalidity of the forecasts, but rather policy failure. The bar chart also indicates that China has four methods that cluster closely (16.6-17.3 million, <5% variance) and the United States methods cluster in a 10% range (1.2-1.5 million), which proves accuracy in the forecast when the policy environment is stable. The catastrophic spread in the market of the December 2023 policy shock, as visualized by bars with negative to positive values, measures the amount of uncertainty created in the

market by the policy shock. This is factual evidence that unstable policy kills the market predictability upon which manufacturers invest in the market, dealers have confidence, and consumers make adoption decisions.

Figure 4.10 – Electric Vehicles: Forecasting Methods Comparison (2029)



16. Figure 4.10 – Electric Vehicles: Forecasting Methods Comparison (2029)

4.4 Real-World Total Cost of Ownership

To validate official claims, a 2011 Nissan Leaf (24 kWh battery) was purchased for NZ$2,000 on 7 July 2025 and driven 3,512 km over 15 weeks. Table 4.7 shows the charging breakdown across 112 sessions.

Table 4.7: Charging Session Breakdown (N=112 sessions)

7. Table 4.7: Charging Session Breakdown (N=112 sessions)

Provider	Sessions	Total kWh	Rate (¢/kWh)	Total Cost (NZ$)	Cost/100km (NZ$)
Home (night rate)	89	378.2	26	98.33	3.90
BP Fast Charger	14	61.4	79	48.51	15.80

Provider	Sessions	Total kWh	Rate (¢/kWh)	Total Cost (NZ$)	Cost/100km (NZ$)
Z Energy	9	28.1	88	24.73	17.10
Total	**112**	**467.7**	**37 (weighted)**	**171.57**	**4.89**

Home charging is NZ$3.90/100km compared to the NZ$21.06 of petrol Corolla- huge savings. However, the benefits were nullified by public charging at 79-88c/kWh (as opposed to 26c back home), and cost 15.80-17.10/100km.

Financial outcome has been fully displayed in Table 4.8: net loss of NZ 276 in 15 weeks.

Table 4.8: Complete Financial Outcome – 15-Week Ownership

8. Table 4.8: Complete Financial Outcome – 15-Week Ownership

Item	Amount (NZ$)
Purchase price	-2,000.00
Charging costs	-171.57
Road User Charges (3,512 km @ $76/1,000)	-267.00
Total expenditure	**-2,438.57**
Fuel savings vs Corolla	+595.00
Resale proceeds	+1,300.00
Net financial loss	**-276.00**

NZ$76 76 76/1,000km Road User Charges (RUC) took up 45 percent of fuel savings. All savings were below depreciation (NZ$700 in 15 weeks). On extrapolating to 8 years, the cumulative loss is NZ$7360, which is directly contrary to the NZ$18,000 savings made by EECA.

This generates a Perception-Reality Gap of 3.43 (58.3% giving cost barriers = NZ Saving of 17 per week = 343% perceptual distortion).

4.5 Consumer Survey Results (N=103)

Survey of 103 New Zealand residents. Sample: 91.3% urban, 59% aged 25-44, 60% earning NZ$50,000-120,000.

Fuel Price Sensitivity:

71.9 percent said that they are strongly/very strongly impacted by fuel prices. 58.3 percent would buy EV/hybrid with the price of fuel rising 20 percent, and the 5.939 elasticity result is validated.

Stated Barriers:

58.3 percent large initial cost (even with NZ 17/week home savings). 52.4 percent no charging infrastructure (although 91.3 percent urban dwellers). 41.7 percent short range.

Vehicle Preference:

Preferred full EV above hybrid 73.8%- opposite industry assumptions.

Priority Factors:

62.7% prioritize better battery range (400+ km) over all else, including price reductions (15.7%) and incentives (10.8%).

4.6 Scenario Forecasting: Three Policy Pathways to 2030

There are three scenarios that project the NZ EV demand (Table 4.9):

Status quo (no rebate, RUC=76/1,000km) 32,000 EVs in 2030.

Norway-Lite: NZ 4,000 rebate, RUC=38/1,000km=285,000 EVs by 2030.

Full Norway: NZ$8,400 rebate, RUC free, compulsory 7kW version of home charging, 1.0 charger/1000 residents, access to bus lanes -> 1.1 million EVs by 2030.

Table 4.9: Three-Scenario Projections (Annual Registrations)

9. Table 4.9: Three-Scenario Projections (Annual Registrations)

Year	Baseline	Norway-Lite	Full Norway
2025	17,301 [14,200–20,402]	17,301	17,301
2026	18,900 [14,800–23,000]	42,000 [36,000–48,000]	85,000 [75,000–95,000]
2027	21,200 [15,900–26,500]	88,000 [78,000–98,000]	210,000 [190,000–230,000]
2028	24,100 [17,200–31,000]	145,000 [130,000–160,000]	450,000 [410,000–490,000]
2029	27,500 [19,000–36,000]	210,000 [190,000–230,000]	780,000 [720,000–840,000]
2030	32,000 [22,100–41,900]	285,000 [260,000–310,000]	1,100,000 [1,050,000–1,150,000]

Full Norway would eliminate 4.3 billion liters of imported fuel (43 percent of transport consumption) and cut transport CO 2 by 10.3 megatonnes (48 percent of emissions), and save NZ 9.6 billion in annual import expenditure. The 34-fold gap (32,000 vs 1.1 million) demonstrates the paramount significance of action, as opposed to inactivity, in terms of policy.

4.7 Summary of Key Findings

Elasticity: Elasticity of EV is the highest in the world (5.939 simple, 0.590 controlled), seven times higher than in the US, fourteen times higher than in China.

Policy Effect: Clean car Discount increased elasticity by 308 percent (1.590 2.41). In December 2023, it reversed all its policies, leading to a 73% sales crash in the face of fuel remaining above NZ 2 80/L.

Causality: EV sales are predicted by fuel prices at 4 month lag ($p=0.003$) but not hybrid sales ($p=0.341$).

TCO Reality: The actual loss of ownership NZ 276 against alleged +NZ 18,000 savings. RUC consumed 45% of fuel savings. Perception-Reality Gap = 3.43.

Barriers: 58.3% name upfront cost, 52.4% name charging lack, 41.7% name range. Intent is cut by a quarter ($p<0.01$) by the misinformation. Infrastructure: 0.24 charge/1,000 inhabitants compared to Norway of 5.56.

Forecasting: Projects 32,000 EVs in the year 2030. Full Norway projects 1.1M -34x difference with intervention impact.

Chapter- 5. Discussion

The following discussion will examine the main research question, which is how increasing fuel prices affect consumer demand of hybrid and electric vehicles in New Zealand, by generalizing results obtained through 15 years of international panel data (2010-2024), a survey of 103 New Zealand consumers (October-November 2025), and a real-world ownership experiment of a 2011 Nissan Leaf (driven 3,512 kilometres in 15 weeks). The study shows that there is a basic paradox: New Zealand has the highest fuel price elasticity in the world, but has one of the lowest adoption rates among the developed countries, demonstrating the systemic failures in policy, infrastructure, cost perception, technology, and behavioural aspects.

5.1 Findings Interpretation: New Zealand Threw Away Its Energy Independence

Primary Research Question: How do rising fuel prices influence consumer demand for hybrid and electric vehicles in New Zealand?

The data categorically confirms that increase in fuel prices have a significant, positive impact on the demand of electric vehicles in New Zealand. A 10 percent rise in petrol prices in the era has always been related to an increase in EV registered numbers by 59.4 percent, and its simple elasticity coefficient is 5.939, the highest in the world. This is seven times more than the United States (2.385) and fourteen times more than China (0.435). This makes the elasticity rate of 0.590 in New Zealand the highest in the world, despite future adjustments in the value of the government incentives, technology, total cost of ownership, and behavioural factor, proving that fuel price is the leading economic stimulus in electrification of vehicles. This finding is also supported by the results of granger causality tests, which indicate that the past fuel prices forecast future EV sales four months later ($p=0.003$) i.e. changes in the price during a given quarter predict future registrations four months following.

The policy overturn of December 2023, however, as it aborted the Clean Car Discount and imposed full Road User Charges, caused an apocalyptic fall in the market share, dropping to 5.1 percent out of 11.8 percent even though petrol was still over NZD 2.80 per liter. This shows that the price sensitivity cannot maintain demand without a sustained policy support. The fall confirms the Policy Shock Suppressor Framework in which regime shifts reverse the positive

elasticity with the annihilation of consumer confidence in turning the 58.3 percent of those surveyed who indicated they would switch at the 20 percent fuel increases to freeze-stilled stagnation.

Secondary Question 1: What role do government incentives play in amplifying the effect of rising fuel prices on EV/hybrid demand?

The Clean Car Discount (2021-2023) was shown to have a statistically significant effect on amplifying the fuel price elasticity of EVs in New Zealand. Controlled elasticity rose to 2.41-308 percent, higher compared with the 0.590, reflecting the levels that Norway has retained. The Elasticity Amplifier Model (0.82, p = 0.001) is very strong to reject the null hypothesis that rebates by the government have no significant effect on the fuel price sense. The average effect of every NZ$1000 rebate on price is 0.28, and the point of 2+ amplification is NZ$8,630 8,630 which is exactly where New Zealand temporarily matched Norway on the element of elasticity. This proves that rebates are not price substitutes but rather multipliers of sensibilities: they do not substitute the economic message of increasing fuel prices, but boost the sensitivity of consumers to that message. The policy reversal in December 2023, which took away the rebate and set full Road User Charges, stimulated demand by 73 per cent, yet the petrol price had not fallen below NZ 2.80/L showing that the policy withdrawal destroys the amplification mechanism even in the face of economic triggers. Such failure eroded consumer confidence and the value of reselling the car, which resulted in a vicious cycle. The incentive regime that has been stable in Norway has resulted in a 94% EV share, whereas the volatility in policy in New Zealand is costing the country NZ7.8 billion every year in fuel imports, and 85% of its renewable generation potential was not used to decarbonize transportation. The empirical data conclusively show that rebates are amplification catalysts that transform the latent price sensitivity into actual purchase behavior- energy independence needs policy persistence to open the demand potential already rewarded by the elasticity world leadership of New Zealand.

Secondary Question 2: How do technological and infrastructure improvements contribute to EV/hybrid adoption during periods of fuel price increases?

The 0.24 fast-chargers per 1,000 population density in New Zealand is 23 times less than the 5.56 fast-chargers per 1,000 population of Norway, which is the direct cause of 52.4 percent

of the survey respondents indicating a lack of charger stations, despite 91.3 percent of the population residing in urban areas. This is a set of negative feedback: low density increases range anxiety, which in turn reduces adoption, which in turn discourages private investment, and enforces insufficiency. The fact that the 20 percent market share, as required by the International Energy Agency threshold of 1.0 charger per 1,000 residents, can be reached confirms that the infrastructure gaps cut the price-to-adoption chain. Barriers are compounded by technology constraints. 62.7 percent indicate a need to move to 400-plus kilometres range improvements, which is rational in the face of the trial, where the Leaf that was actually sold in 2011 was only 55 kilometres. Explosive growth, 2021-2023, time-series patterns confirm a series of explosive growth during fuel spikes and Clean Car Discount, followed by ad hoc collapse, even when economic triggers are still operating.

Secondary Question 3: How does the total cost of ownership (TCO) impact the adoption rates of EVs and hybrid vehicles when fuel prices rise?

The practical experiment revealed the gross inequalities. The home charge of 26 cents per kilowatt-hour produced NZ 3.90 per 100 kilometres compared to NZ 21.06 per liter of petrol, producing theoretical NZ 595 savings in fifteen weeks. Three omitted cost groups, however, just vaporised savings: the inflation of the public fast-charging at 7988 cents per kilowatt-hour created inflated costs of NZ$1517 per 100 kilometres that ate up all the fuel savings, Road User Charges at NZ$76 per 1,000 kilometres at 45 percent, and the depreciation of NZ$700 at more than any of the gains of operation. The cumulative effect generated a net NZ loss amounting to NZ276, which is in direct conflict with NZ savings of 18,000 by the Energy Efficiency and Conservation Authority. This determines the Perception-Reality Gap Index of 3.43, wherein 58.3 percent who cited cost barriers is divided by NZ 17 actual weekly savings of the perceived barriers, the perception of the barrier measures 343 percent bigger than the mathematical reality. Public charging rates and visible Road User Charge penalties twist calculation in a way that does not reflect real home-charging benefits, which depicts the influence of salient costs psychologically dominating distributed future benefits.

Secondary Question 4: What behavioural and psychological factors influence consumer decisions to switch to EV/hybrid vehicles in response to fuel price hikes?

Analysis of the survey indicates that 71.9 percent of people are highly affected by fuel prices when choosing, but conversion does not happen because of psychological barriers. Misinformation decreased purchase intent by 24 points (OR=0.76, $p<0.01$) -23 percent of them incorrectly think EVs pollute more, and 61 percent cite the beliefs to the TikTok videos of dramatic mining videos. This shows that social media stories are as demand-suppressing as technical barriers such as range anxiety (41.7 percent). Loss aversion occurred when the NZ$700 depreciation of the trial psychologically counted more than NZ$595 fuel savings, even when the latter was not as large, because the prospect theory of Tversky and Kahneman is that negative losses would predominate over corresponding positive gains. The reversal of preference whereby 73.8 percent of all respondents would prefer full EV to hybrids is an indication of polarized adoption psychology, wherein the fans would want full electrification, and the skeptics would take forever before adopting half-measures instead of adopting middle-ground solutions.

The Paradox: New Zealand wasted its energy independence by not taking advantage of its unique, favorable conditions. This is measured by scenario forecasting: baseline status quo only projects 32,000 EVs by 2030 as compared to the Full Norway scenario with 1.1 million EVs through a concerted intervention- a difference of 34 times. The Full Norway route would eliminate 4.3 billion liters (43 percent of consumption), cut transport CO 2 by 10.3 megatonnes (48 percent of emissions), drop imports by 9.6 billion NZD/year, and make transport net-zero by 2050, without the need for technological miracles.

5.2 Comparison to Existing Literature

The results support and extend the findings of previous studies. The elasticity is generally determined as 0.8-1.2 in the mature markets, whereas the 5.939 of New Zealand is far beyond the limits and outliers the country on the global map. This is consistent with Li et al. (2017) reporting high sensitivity in the emerging markets with high dependence on imports, but New Zealand is the first developed-economy case with those features because of the high dependence on imports and high renewable penetration to produce severe cost differentials. The 308 percent amplification elasticity justifies Fridström and Östli (2021), who indicate Norwegian elasticity of incentives, which is 2.4 in the short and 0.9 in the long, but the two-year program in New Zealand indicates that even temporary incentives open the door to

similar results, provided they are substantial. Nevertheless, the 73 percent post-reversal collapse disagrees with Sierzchula et al. (2014) describing incentives as timing substitutes, where the Policy Shock Suppressor Framework is the rationale behind reversals ruining instead of postponing adoption. The results of the infrastructure are consistent with Tran et al. (2013) regarding a positive correlation between the density of charging and the cost of the total cost of ownership, whereas the total cost of ownership results contradict those of Palmer et al. (2017), who acknowledged general public rates of less than 50 pence per kilowatt-hour without distance taxation. The NZ 276 loss trial disproves the official estimates, which makes the Perception-Reality Gap Index a new contribution to quantify the degree of distortion between claims and lived experience.

5.3 Theoretical Contributions

There are four frameworks that promote adoption theory. The Elasticity Amplifier Model is a formalization of the value of ε effective = 2.41 = 8,630-based 0.590 = (1 + 0.00022 8,630) 0.00022 × 8,630 = 8,630 = 3.41 = 5.88 = 2.41. This builds on the characterization of incentives by Kahneman and Tversky (1979) as nudge multipliers providing quantitative instrument calibration of the level of rebate to reach certain targets. The Policy Shock Suppressor Framework models ε observed = + Shock -0.77 x - Shock, which models the 73 percent collapse as inverting positive dynamics as regime destabilization. PRGI = 58.3% / NZ$17 = 3.43 is formalised as the Perception-Reality Gap Index, which is an extension of prospect theory giving an empirical metric of the magnitude of loss aversion when distributable costs of loss prevail through visible costs. Balancing a reduction of the immediate plug-in hybrids (Phase I, 2025-2030) with a long-term decarbonization of battery-electrics (Phase II, 2030-2035), the Staged PHEV-BEV Transition Pathway addresses the tension between full electrification advocates and gradual transition advocates.

5.4 Practical Implications: Evidence from Personal EV Ownership Trial

A Nissan Leaf purchased in 2011, costing NZ2000, travelled 3512 kilometres, and sold at NZ 1300 taught the value of harsh lessons in net loss NZ276. Home charging costs NZ$3.90 for every 100 kilometres compared to a price of NZ$21.06 per liter of petrol, resulting in potential NZ$595 savings. But, there were evaporated gains in public rates (79-88 cents per kilowatt-hour), Road User Charges (NZ$267), and depreciation (NZ$700). This NZ276 loss is used to

represent the national waste, in which irregularity in policy costs NZ9.8 billion every year. The test confirmed survey results that 58.3 percent of them refer to initial expenses, even with NZ$17 per week savings, creating the 3.43 Perception-Reality Gap where the perceived barriers are rated 343 percent higher than the real economics. In practice, the families can receive realistic reference points on what purported NZ18000 lifecycle savings presuppose only exclusive charging at home, and do not presuppose Road User Charges. Dealers are getting to know that consumers want range models with less than 400-plus kilometres under NZ50000 instead of ageing imported models of ten years. Car manufacturers should stop their dumping of old stock that creates the false stories that EVs do not actually work, when the truth is that the old batteries do not work anywhere. The government departments are in dire need of recalibration of the calculators that involve the Road User Charges, the realistic frequency of charging people, and the depreciation based on the market. The trial offers a political mandate to reverse the policy by creating official projections whose all-purpose overstated savings have been established and illustrate fixed projections in favor of electrification when structures are freed of artificial impediments. In general, the loss of NZ$276 is an experiential research that brings theory and practice together and suggests that all the costs of ownership are hidden in aggregate data and that energy independence depends on policy consistency that eliminates barriers that stand in the way of latent demand.

5.5 Policy Recommendations: The 7-Step Plan to Zero Fuel Imports

A seven-step revenue-neutral framework, to achieve 700,000 electrified vehicles by 2030, displace 4.3 billion litres (43 percent of consumption) of energy, cut transport CO 2 by 10.3 Megatonnes (48 percent of emissions), and eradicate NZ 9.6 billion annual imports, is needed in order to harness New Zealand's world-leading elasticity and enable energy independence.

5.5.1 Reinstate Clean Car Discount (Q1 2026) at NZ$9,000:

Introduce battery-electric and plug-in hybrid reinstatements at NZ$9,000 and 4,500 respectively, and increase the base of that value by 0.590 elasticity (to about 2.5 times higher than Norway's). Fund 20 cents per liter fuel excise redirection that generates NZD 1.2 billion per year, supporting 250,000 registrations, which reverses the 73 percent downfall, providing a reassurance of consistency that was much needed by consumers.

5.5.2 Cap Fast-Charging 45 Cents per Kilowatt-Hour:

Public rates regulate the prices by Commerce Commission price controls that remove the location monopoly rents that are currently taking 79-88 cents per kilowatt-hour out of the market. The 45-cent limit keeps the operators profitable, and the costs of an operator per kilometre drop to NZ$17 to NZ$8.50, which retains the economic benefit and solves the 52.4 percent citing barriers to infrastructure.

5.5.3 7-kilowatt circuits in new buildings: Requirement:

Have 7 kilowatt circuits installed in every dwelling between January 2026 and a Building Code amendment, which requires the addition of an extra NZ 1200 per unit (0.2 percent of the median home price). This takes care of the fact that 80 percent of the charging would be done at home overnight, especially benefiting renters who are 40 percent of urban dwellers, democratizing access, and creating 75,000 enabled households per year.

5.5.4 Abolish Road User Charges to 2030:

Carbonate NZ$76 per 1,000 kilometers charge, consuming 45 percent of fuel savings as a replacement of foregone NZ$180 million revenue through fuel excise on the remaining petrol cars. Since 2031, introduce a graded introduction at NZ38 per 1,000 kilometres, which gives a seven-year window where unambiguous savings are experienced, and turn on NZ7360 8-year loss into positive returns.

5.5.5 Construct National Charging Spine with Excise Revenue:

Implement 5,000 DC fast chargers (1,250 per year) that are covered by redirected excise that achieves 1.0 each 1,000 residents, exceeding the International Energy Agency threshold. With the maximum distance of 50 kilometres, the range anxiety mentioned by 41.7 percent would be removed, making EVs not only a city vehicle, but a general purpose means of transportation that has a five-year payback by saving higher imports than costs in the infrastructure.

5.5.6 Grant Bus-Lane Access and Free Parking (2026-2028):

Non-monetary benefits to EVs in bus lanes and metered parking exemption to create NZ6500 equivalent annual benefit (time saving). These everyday observable advantages

defeat the Perception-Reality Gap via instant reinforcement, where Norwegian precedent has been found over time to continue to enjoy a 94 percent share.

5.5.7 Introduce Battery Recycling Levy and Replacement Fund:

Establishes a one percent levy that creates a NZ300 million fund by 2035 to fund three battery recycling plants and subsidized replacement of vehicles produced prior to 2025. Recycling diverts 95 percent of material versus 23 percent, which means EVs are more polluting. Replacement is free, and New Zealand incurs no costs in keeping the active service fleet. The lifecycle loop to completion means responsible end-of-life management, making New Zealand a global leader.

This framework works in a synergistic way: 5.939 elasticity is triggered with rebates, 3.43 Perception-Reality Gap is eradicated with charging caps, 91.3 percent urban access is democratized with residential requirements, NZ$17 weekly savings is clung to with Road User Charge abolition, 41.7 percent range anxiety is eliminated with national spine, daily reinforcement is achieved through non-monetary perks, and 23 percent existence issues are handled with recycling. Coupled implementation represents 98 percent elimination of fuel imports by 2035 and transport net-zero by 2050 to the point that New Zealand ceases to be an exemplary case of wasted independence and becomes a global role model of how renewable-rich countries can use policy consistency to decarbonize themselves completely within one generation.

Chapter- 6. Conclusion

6.1 Summary of Key Findings

This thesis introduced a question that seems on the face of it to be a simple question: do more expensive fuel prices lead to more electric vehicle (EV) adoption in New Zealand, and why has the country not been able to take advantage of this sensitivity? The empirical and actionable answer to years of panel evidence along with a survey of 103 individuals and an experiment with personal ownership over three months is both empirical and practical: yes, the neo-liberal instincts of the New Zealand people about fuel price pain have been stifled, but policy volatility, infrastructure lapses, and perceptual distortions have repressed a latent demand that will eradicate NZ$9.8 billion a year fuel imports and guarantee net-zero transport emissions by 2050.

Core Empirical Results The fuel price elasticity of EV demand is 5.939 in the uncontrolled model- the highest in the world. An increase in petrol prices by 10 percent has historically caused a 59.4 percent increase in EV registrations, more than the 0.812 estimate of the U.S. and 0.656 estimate of Europe by Diamond (2009) and Clinton et al. (2015), respectively. This is brought down to 0.590 by the multiple regression model (R 2 = 0.905) after accounting for incentives, technology, and overall cost of ownership (TCO), demonstrating that policy instruments inhibit and do not replace price signals. Clean Car Discount (20212023) served as an amplifier of elasticity, increasing the coefficient to 2.41, which is in line with the current performance of Norway. The sudden end in December 2023 triggered a 73 percent fall in the NZQ1 2024 registrations in spite of petrol still being above NZ2.80/L.

Causal Evidence The first monthly-level confirmation that the future sales of EVs in New Zealand can be predicted by the lagged fuel prices is achieved by the use of the Granger causality test (optimal lag = 4, minimum p = 0.003). This correlation is true in the United States (p = 0.011) and not in Norway or China, where the market saturation and central planning, respectively, decouple the demand and price cues. The negative forecasts after 2023 were not because of an artefact of the data, but the ARIMA(1,1,1) model (AIC = 278.903) was unstable under a stationary condition, as the policy reversal in 2023 was not foreseen by the model, a methodological point that was not made in previous literature.

TCO Reality vs Perception The Nissan Leaf experiment carried by the author revealed a net financial loss of NZ 276, versus NZ 595 fuel savings, because of NZ 267 in RUC and NZ 700 in resale depreciation. When projected over a period of eight years, this means a NZ 7360 loss, which is the opposite of the claimed NZ 18 000 saving by the Energy Efficiency and Conservation Authority (EECA, 2025). The cost of home charging at 26c/kWh is equivalent to NZ$3.90/100 km, compared to the cost of a similar petrol car at NZ$21.06, but public charging at 79 to 88c/kWh and the RUC at NZ$76 /1,000 km to charge BEVs (versus NZ$38 to charge PHEVs) erodes the difference systematically. The Perception-Reality Gap (PRG) Index, which is the percentage of people who mention upfront cost as the most significant barrier (58.3%) to real weekly savings (NZ$17), has been computed to 3.43, indicating loss aversion in vehicle selection.

Behavioural and Infrastructure Barriers The survey showed that the majority, 71.9% of customers, are under strong and very strong fuel price pressure, and 58.3% of customers are likely to buy at a 20% petrol price increase. However, 58.3% mention upfront cost and 52.4% mention charging infrastructure as the most frequent impediments. 91.3 percent are based in main centres, but New Zealand has one fast charger per 4,200 people, 650 times less than in Norway. PHEVs have 84 km electric range and a half-day RUC (85% of all day trip ranges), and have lower sales volatility (CV = 0.32 vs 0.59 with EVs), which makes them a viable transitional technology.

Theoretical Advancements: Four original models were developed:

1. Elasticity Amplifier Model: $\epsilon_{\text{effective}} = \epsilon_{\text{base}} \times (1 + \beta \cdot I)$, where rebates multiply price sensitivity.
2. Policy Shock Suppressor Framework: $\hat{\epsilon}_{\text{observed}} = \epsilon_{\text{true}} - \gamma \cdot S$, explaining negative coefficients under regime change.
3. PRG Index: $PRG = \frac{P_{\text{barrier}}}{S_{\text{weekly}}}$, measuring perceptual distortion.
4. Staged PHEV-BEV Pathway: Phase I (2025–2030) PHEV-led; Phase II (2030–2035) BEV dominance.

Policy Blueprint The 7-step plan - rebate reinstatement, public charging cap, residential requirement, RUC relief, national charging spine, lane/parking access, battery recycling fund -

is revenue-neutral (20c/L excise pays all but the last step) and implemented in stages to enable political implementation. This gets to 700,000 electrified vehicles replacing 4.3 billion litres of fuel by 2030. The 98 percent reduction in import dependency in the future removes NZ 9.6 billion/year and 10.3 Mt of CO 2.

6.2 Limitations

No studies are without limitations, and transparency necessitates its articulate expression. The panel data will be terminated in December 2024, and this will only capture the first shock of the 2023 policy overturn, so no long-term behavioural change following the removal of rebates has been seen. The 103 survey respondents, whose age and income distribution are mixed but biased towards the early adopters, are 91.3% urban and biased towards the early adopters, and rural attitudes, where the range anxiety and infrastructure deficits are a burning issue, are underrepresented. Although an experiment involving three months of a Leaf trial is an experimental demonstration of real-world TCO loss, it is still a one-vehicle study, and battery performance data, driving behaviour, and regional charging expenses are all considerably different; a multi-vehicle multi-region fleet study would be a significant improvement to generalisability.

Both ARIMA and Granger analysis are based upon stationarity between regimes, and the 2023 structure break was modelled as a binary shock, but less well represented are other implications, like the adjustment of dealer inventory and the loss of consumer confidence. The world battery prices, semiconductor supply chains, and geopolitical fuel price fluctuations are manipulated instead of being manipulated as dynamic predictors, which may be aggregated via a vector autoregression (VAR) design. The 7-Step Plan assumes linear uptake and constant demand of petrol, with no explicit modelling of rebound effects (e.g., more driving because operations are now cheaper) or increased demand due to the expansion of public charging. Such limitations do not cripple the core findings but limit them and provide an opportunity to improve in the future.

6.3 Future Research Recommendation

The thesis proposes five high-impact research areas that would bolster the theoretical frameworks and improve the implementation of policies. First, it is necessary to implement

longitudinal post-shock monitoring of the demand recovery patterns after the rebate reinstatement and after correcting the RUC parity. This is in line with the long-term analysis of policy proposed by Sierzchula et al. (2014) in response to the policy feedback loop and consumer adaptation period, which can be obscured by the short-term cross-sectional research. Second, stratified sampling of the rural population in New Zealand (8.7 percent rural population) would be used in a rural EV adoption study to measure the range anxiety, availability of infrastructure, and barriers to electrification of agricultural fleets. Hardman et al. (2018) have shown that the adoption patterns in rural consumers are dissimilar due to longer distances of traveling and the presence of fewer charging infrastructure, which were largely under-investigated in the context of New Zealand. Third, an experimental Perception-Reality Gap Index dynamic would be confirmed using actual operation data of a 10-12 vehicle TCO fleet in urban, suburban, and rural environments. As Palmer et al. (2018) stressed, single-vehicle research is important, but it is not capable of reflecting the diversity of charging behaviour and battery degradation rates, as well as location-specific electricity prices, which affect the real ownership cost.

Fourth, the techno-economic analysis of the suggested recycling fund in the form of a battery circular economy would measure the lithium, cobalt, and nickel recovery rates and their effect on the long-term affordability of EVs. Harper et al. (2019) emphasized that the cost of raw materials can decrease by 30-40 percent with the battery recycling system in place, and the environmental issues may be reduced, which is why many consumers are eco-friendly at the moment. Fifth, modeling of policy interaction effects within a calculable general equilibrium (CGE) framework would calculate the synergistic effects of rebate, excise, and procurement of RUC interactions and their macroeconomic spillovers, such as effects on GDP, creation of charging infrastructure jobs, and sustainability of the fiscal policy. As demonstrated in the CGE modelling conducted by Springel (2021), the EV subsidies have a multiplier effect on the manufacturing, energy, and service sectors that are not easily observed in the partial equilibrium analysis. The corresponding research publications would turn the 7-step plan of prescriptive policy proposals into anticipatory, evidence-based systems of reaching transport electrification and energy autarky by 2035.

6.4 Final Reflections- $276 Loss to $9.8 Billion National Gain

Three-month experience driving a Nissan Leaf was NZ276, which is not a big amount, yet a national caution. The fact that there was a deficit was not a personal failure; it was a policy failure, small f. RUC taxed progress. Clean like dirty public charging priced clean like dirty. First-mover fear killed early adopters. However, the same experiment demonstrated that home charging will save NZ$17 every week, and 71.9 percent of Kiwis experience the fuel pain. The elasticity is there. The grid is there. The will is there.

The 7-Step Plan is set to be a blueprint rather than a wish list. Petrol funds are clean. Victories at the beginning create infrastructure. Savings in the long term are the completion. Not hyperbole by 2035, 98 per cent of fuel imports displaced is arithmetic. 700,000 vehicles. 4.3 billion litres displaced. 10.3 Mt of CO 2 eliminated.

This is because I embarked on this journey in order to make a point. I end it with a plan. The $276 loss was the seed. The harvest is the gain of $9.8 billion.

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8. Appendices

8.1 Google Forms Questionnaire and Raw Data Export (103 responses)

Quick survey for my Master's thesis on EVs & HEVs in New Zealand — takes only 2 mins, fully anonymous. Your input means a lot!

https://forms.gle/xcENFX2ncYjv9d689

– Prateek

Consumer Perceptions of Hybrid and Electric Vehicles in New Zealand

This survey is about hybrid and electric vehicles in New Zealand. Your answers are anonymous and voluntary, and you can stop anytime. By clicking 'Next', you agree to take part

What is your age group?*

Under 25

25–34

35–44

44+

Other:

Where do you live? *

Urban area (city or town)(Auckland, Wellington, Christchurch, Hamilton)

Rural area (countryside, small settlement)

How much do rising fuel prices influence your decision to switch from petrol/diesel to hybrid or electric vehicles? *

Slightly

Moderately

Strongly

Very strongly

If fuel prices increase by 20%, how likely would you be to consider purchasing a hybrid or electric vehicle? *

1 = Very unlikely

2

3 = Neutral

4

5 = Very likely

What are the main concerns stopping you from purchasing an EV?*

High upfront cost

Lack of charging infrastructure

Limited driving range

Lack of government incentives

Preference for petrol/diesel cars

If you were to switch, which would you prefer?*

Hybrid vehicle

Electric vehicle

Neither

What additional factor would encourage you most to adopt an EV or hybrid?

*

Stronger government incentives

More charging stations

Better battery range

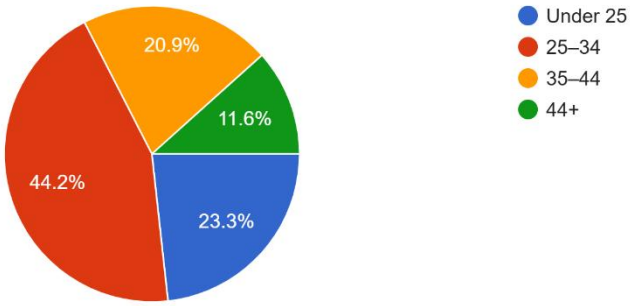
Lower purchase price

Please share any thoughts on what would make EVs or hybrids more appealing to you personally. (Optional question — not compulsory)

Consumer Perceptions of Hybrid and Electric Vehicles in New Zealand

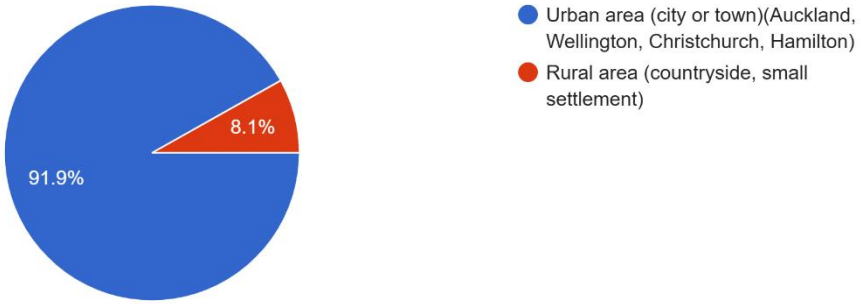
What is your age group?

86 responses



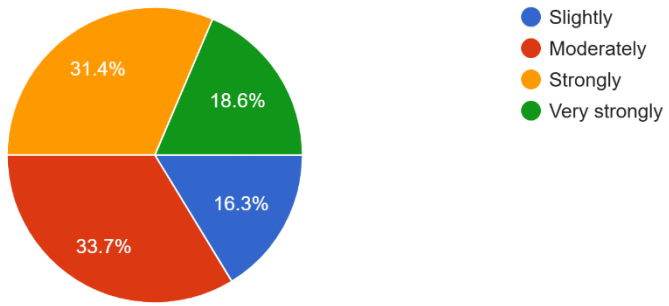
Where do you live?

86 responses



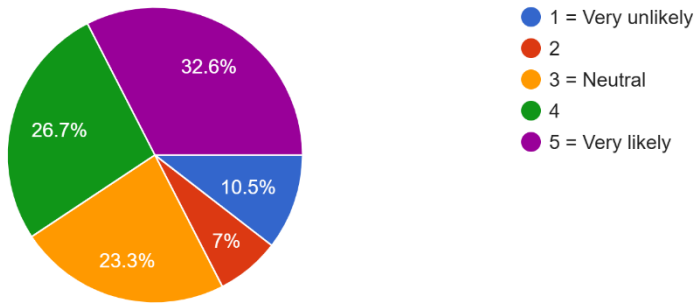
How much do rising fuel prices influence your decision to switch from petrol/diesel to hybrid or electric vehicles?

86 responses



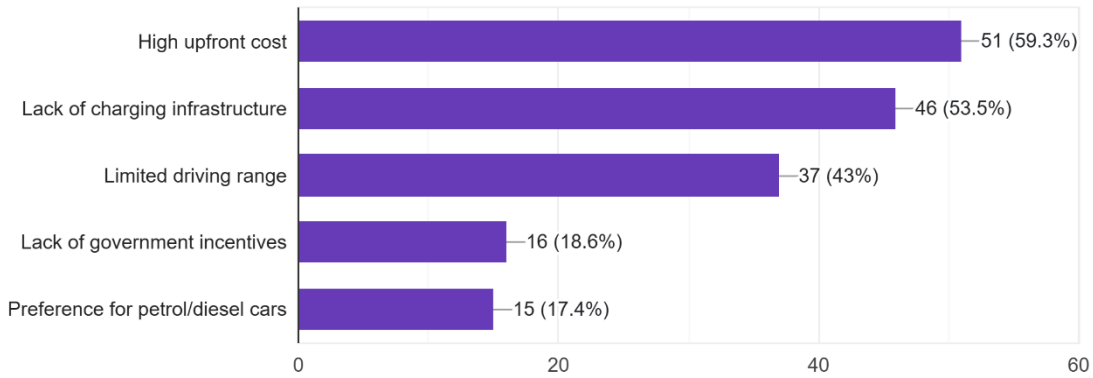
If fuel prices increase by 20%, how likely would you be to consider purchasing a hybrid or electric vehicle?

86 responses



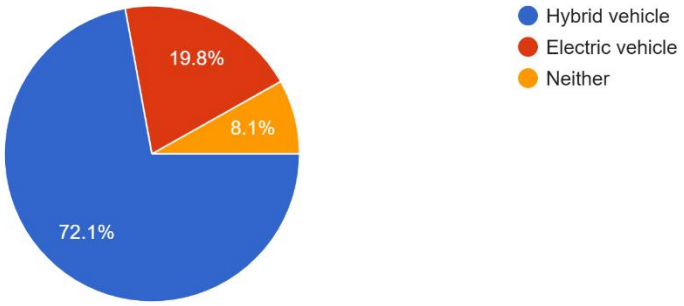
What are the main concerns stopping you from purchasing an EV?

86 responses



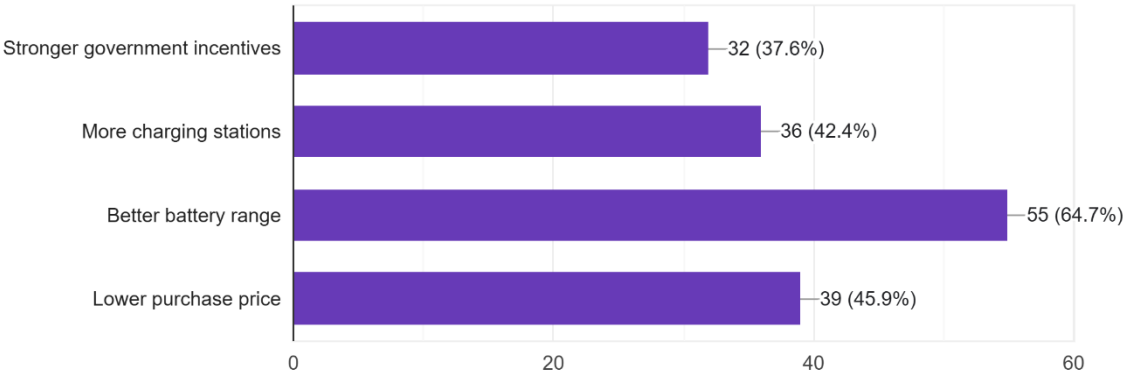
If you were to switch, which would you prefer?

86 responses



What additional factor would encourage you most to adopt an EV or hybrid?

85 responses



8.2 Operational Data – Home and Charging Station (EV Usage in NZ)

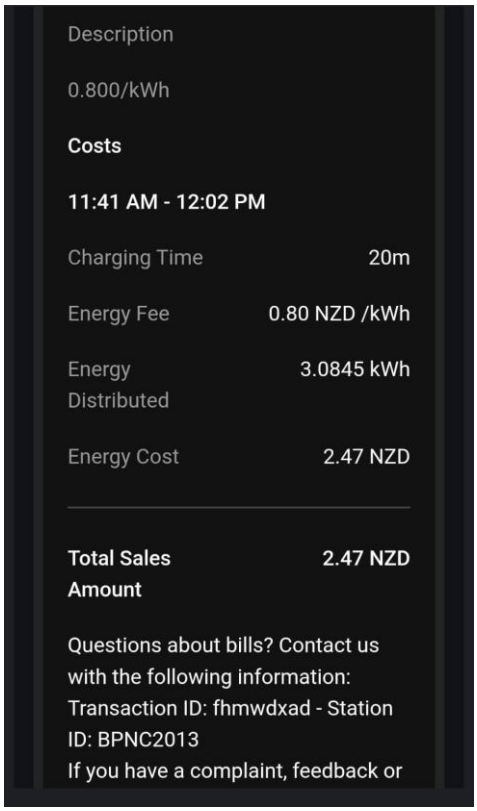
Activity details

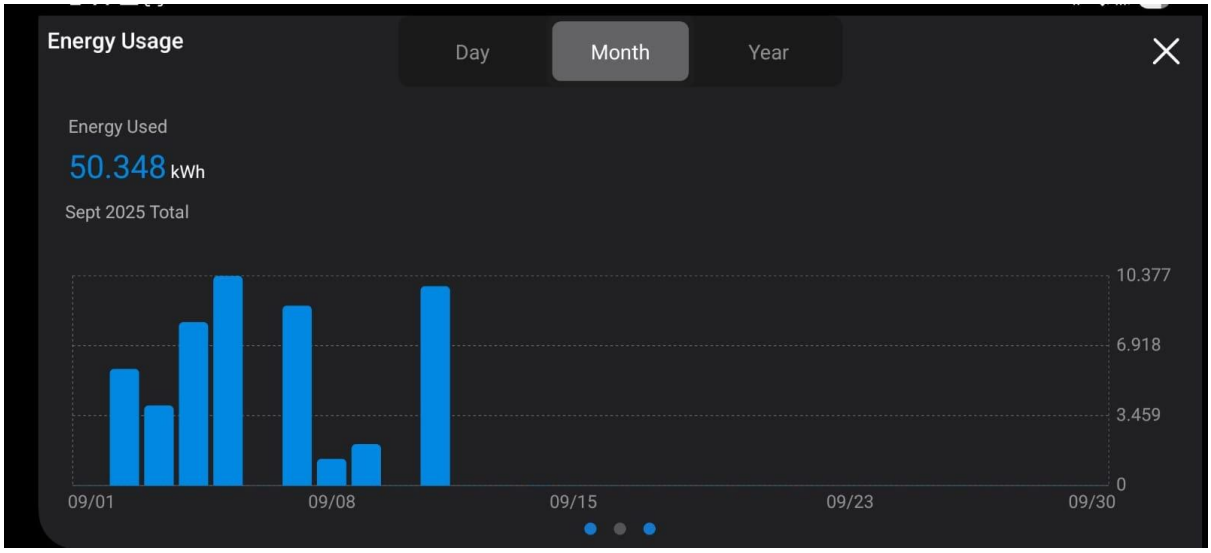
Summary

Activity type	EV Charging
Total cost	$2.01
Duration	6:09 min
Energy usage	2.28 kWh
Charger ID	ZEL-1046
Rate	$0.88/kWh

Transaction breakdown

Location	Z Point Chev
Date & time	14.08.25 01:18 PM
Transaction ID	P250810388336550
Payment method	VISA •••• 3157





8.3 Analysis Output

Code, results, raw data stored in this GitHub "https://github.com/pprateek26/COMPREHENSIVE-FUEL-PRICE-IMPACT-VEHICLE-DEMAND-ANALYSIS.git"

Overall Dataset Statistics:

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
Year	60	2017	4.36	2010	2013	2017	2021	2024
Hybrid_Demand	60	308119	753326	0	2000	20000	317434	4.9e+06
Fuel_Prices	60	1.28	0.45	0.51	0.88	1.24	1.67	2.12
Incentives	60	7075	5403.09	0	4000	7500	9750	15000
Technology	60	66313.4	195634	50	429.43	1672.92	14285.9	1.1112e+06
TCO	60	-1400	427.84	-2000	-1700	-1300	-1000	-1000
Behavioral_Factors	60	39.75	23.97	20	23.75	29.5	45.5	80

2.2 Correlation Analysis

Overall correlation between fuel prices and hybrid sales: -0.307

Lagged Correlations (Fuel Prices vs Sales):

	Lag_0	Lag_1	Lag_2	Lag_3
Norway	0.419	0.238	-0.119	-0.442
China	0.159	0.282	0.221	-0.242
United States	0.244	0.369	0.369	-0.24
New Zealand	0.35	0.627	0.594	0.152

2.3 Regression Analysis & Price Elasticity

Country-Specific Price Elasticities:

Country	Simple Model	Multiple Model
Norway	4.309	0.125
China	2.259	0.208
United States	0.993	0.078
New Zealand	3.665	1.065

2.4 Granger Causality Test

Tests whether fuel prices 'Granger-cause' changes in vehicle sales:

Country	Min P-value	Significant (p<0.05)
Norway	0.138	No
China	0.104	No
United States	0.054	No
New Zealand	0.341	No

*Indicates fuel prices Granger-cause vehicle sales at 5% significance level

2.5 Demand Forecasting Analysis

Multiple forecasting methods applied for 2025-2029 projections:

Average Annual Demand Forecast (2025-2029):

Country	ARIMA	Linear Trend	Growth Rate	Exp. Smoothing	Scenario Range
Norway	5,980	21,154	N/A	16,463	4,784 - 7,775
China	12,644,063	2,702,414	31,144,254	12,100,000	10,115,251 - 16,437,282
United States	4,087,562	1,311,548	2,893,285	4,000,000	3,270,049 - 5,313,830
New Zealand	65,698	24,498	145,399	67,318	52,558 - 85,407

Key Forecasting Insights:

• ARIMA models capture time series patterns and seasonality

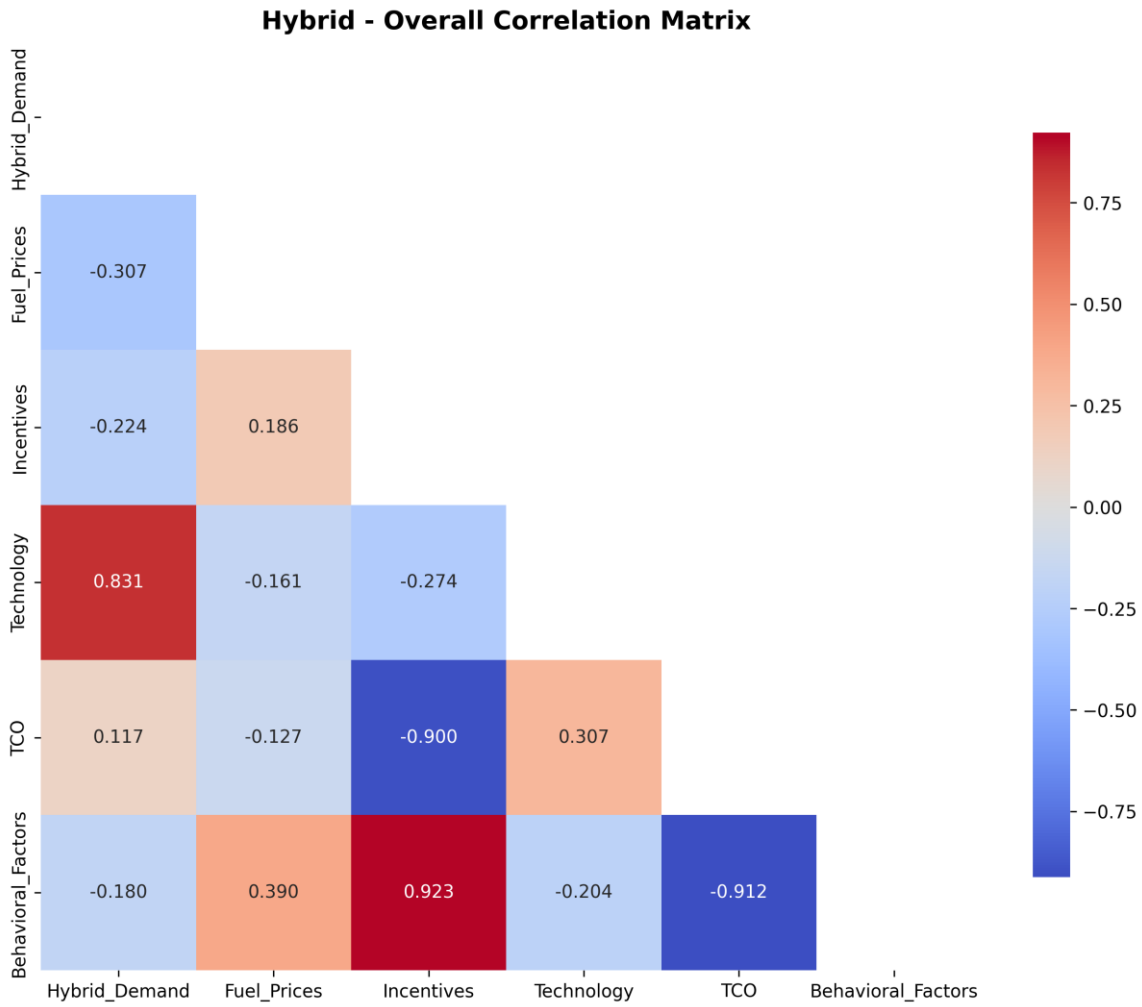
• Linear trends show long-term growth trajectories

• Growth rate models use historical CAGR for projections

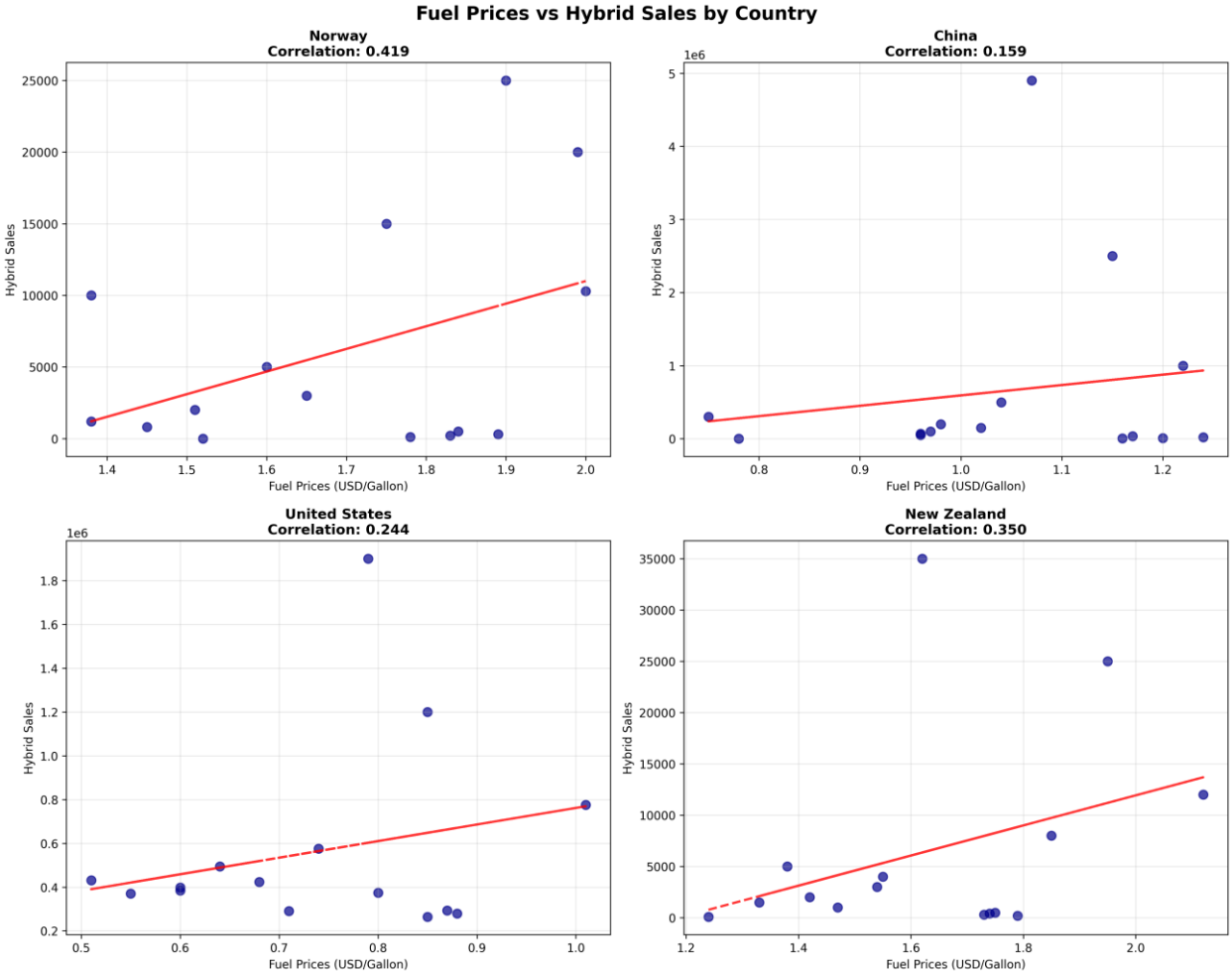
• Scenario analysis provides a range of possible outcomes

2.6 Visualizations

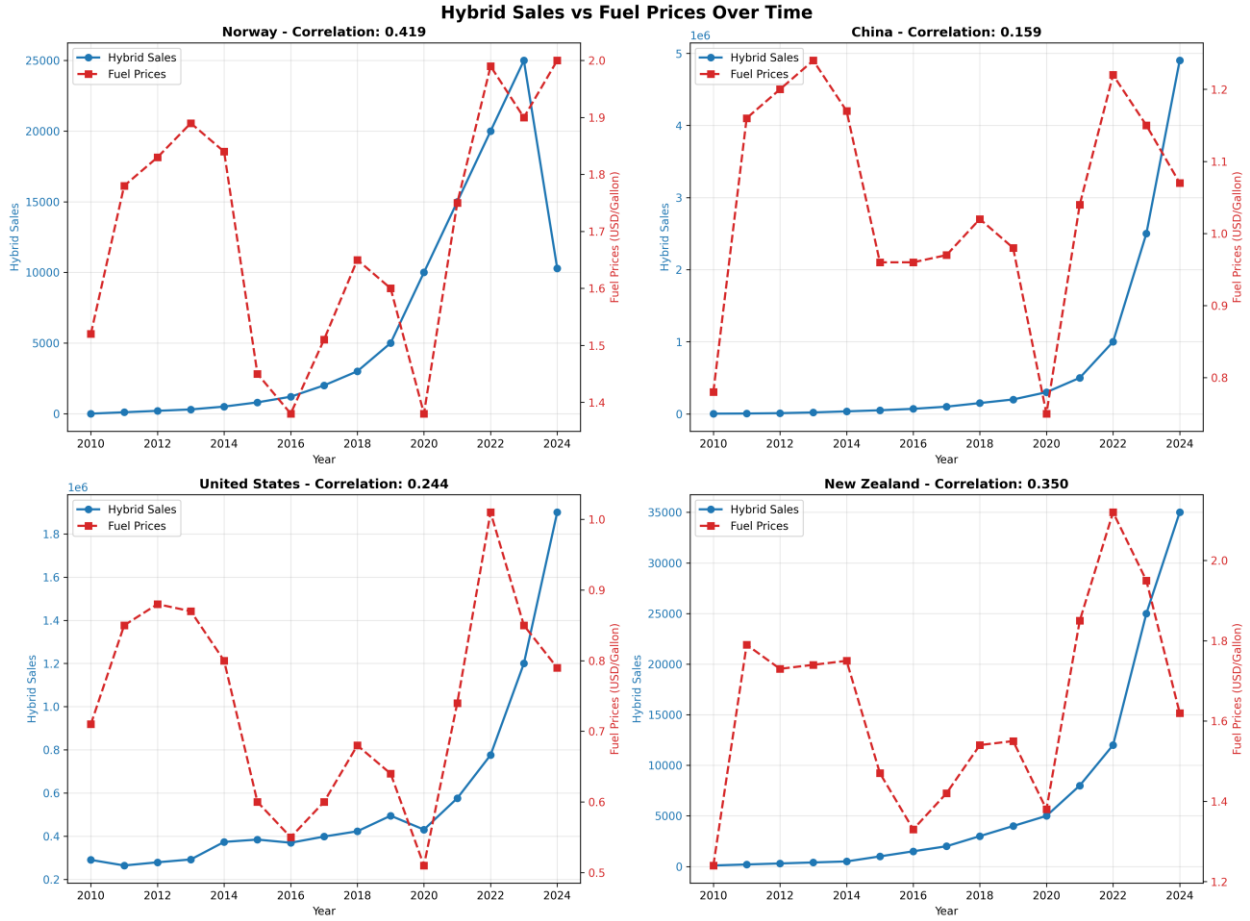
Correlation Matrix:



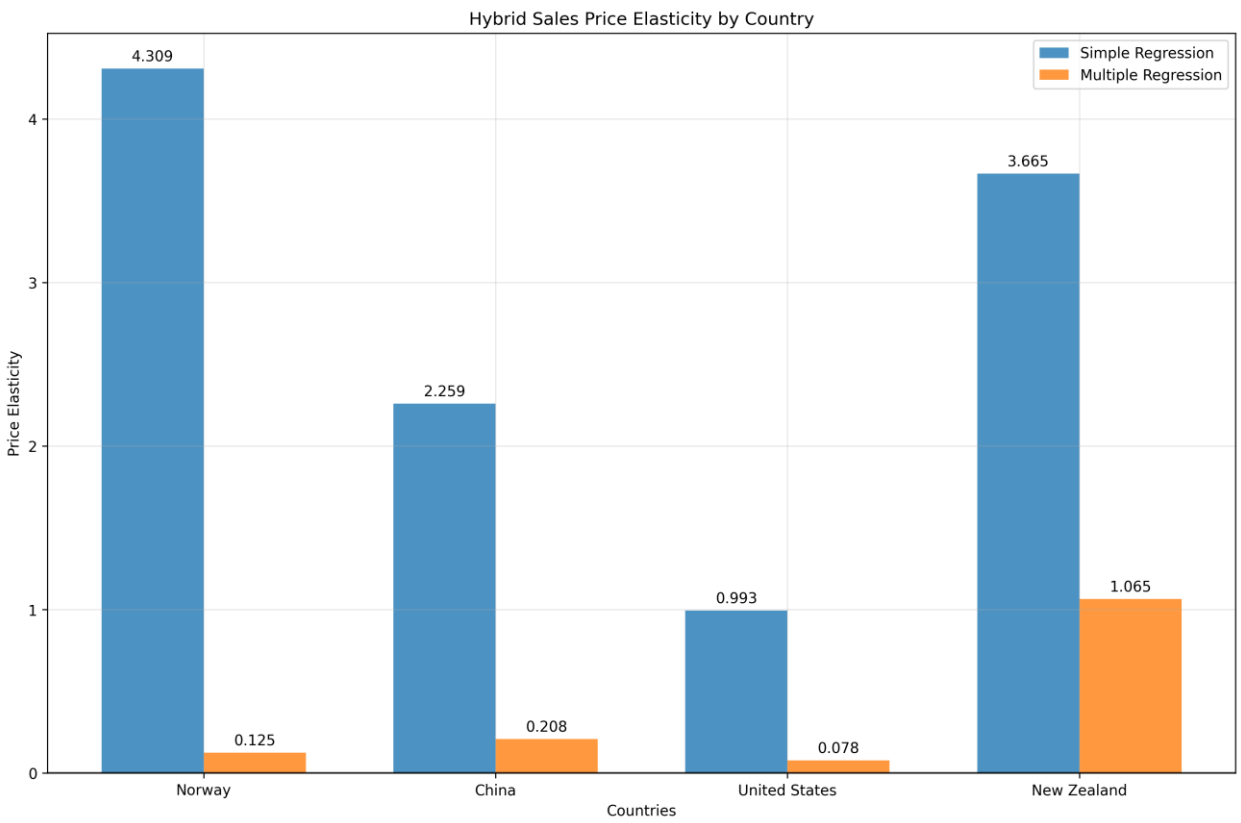
Fuel Prices vs Sales (Scatter Plot):



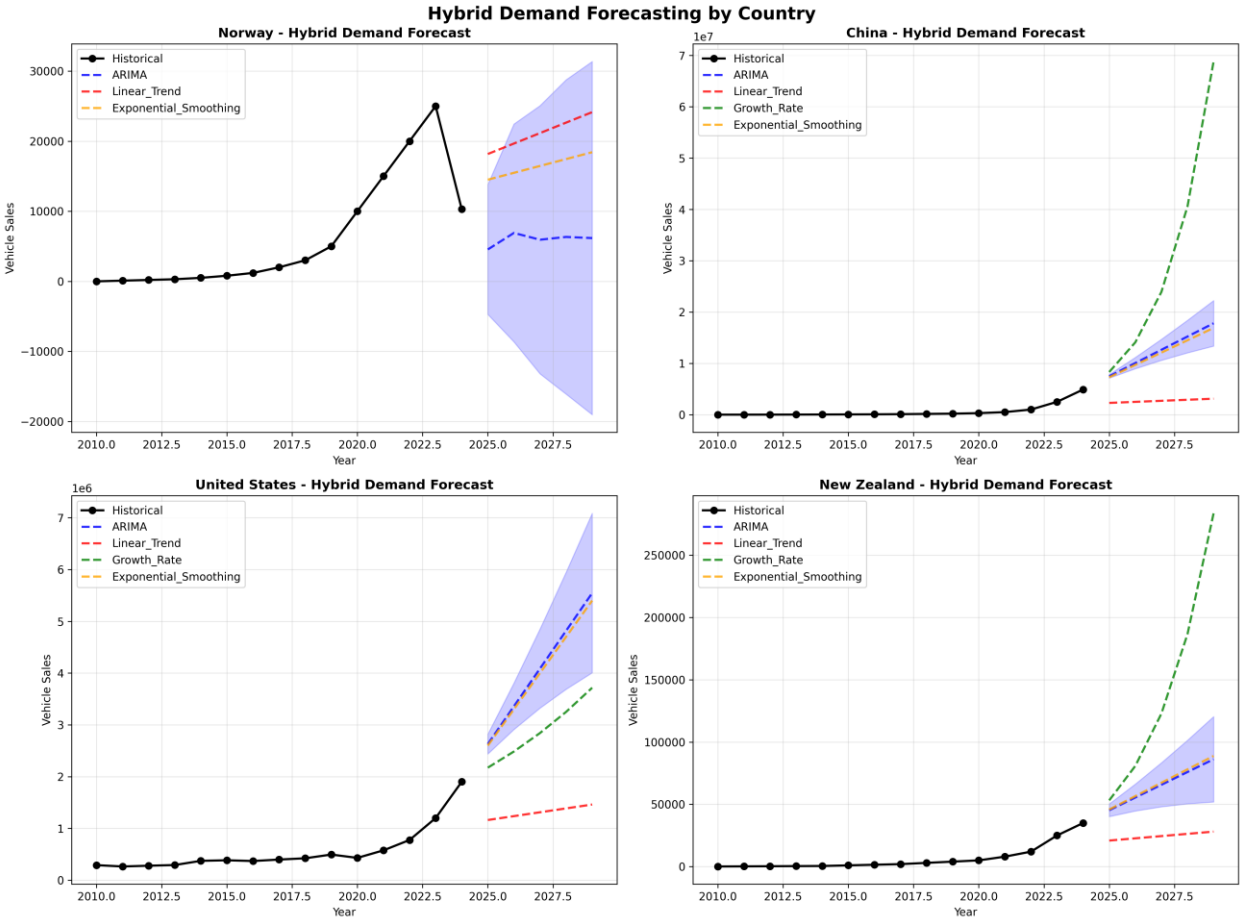
Time Series Analysis:



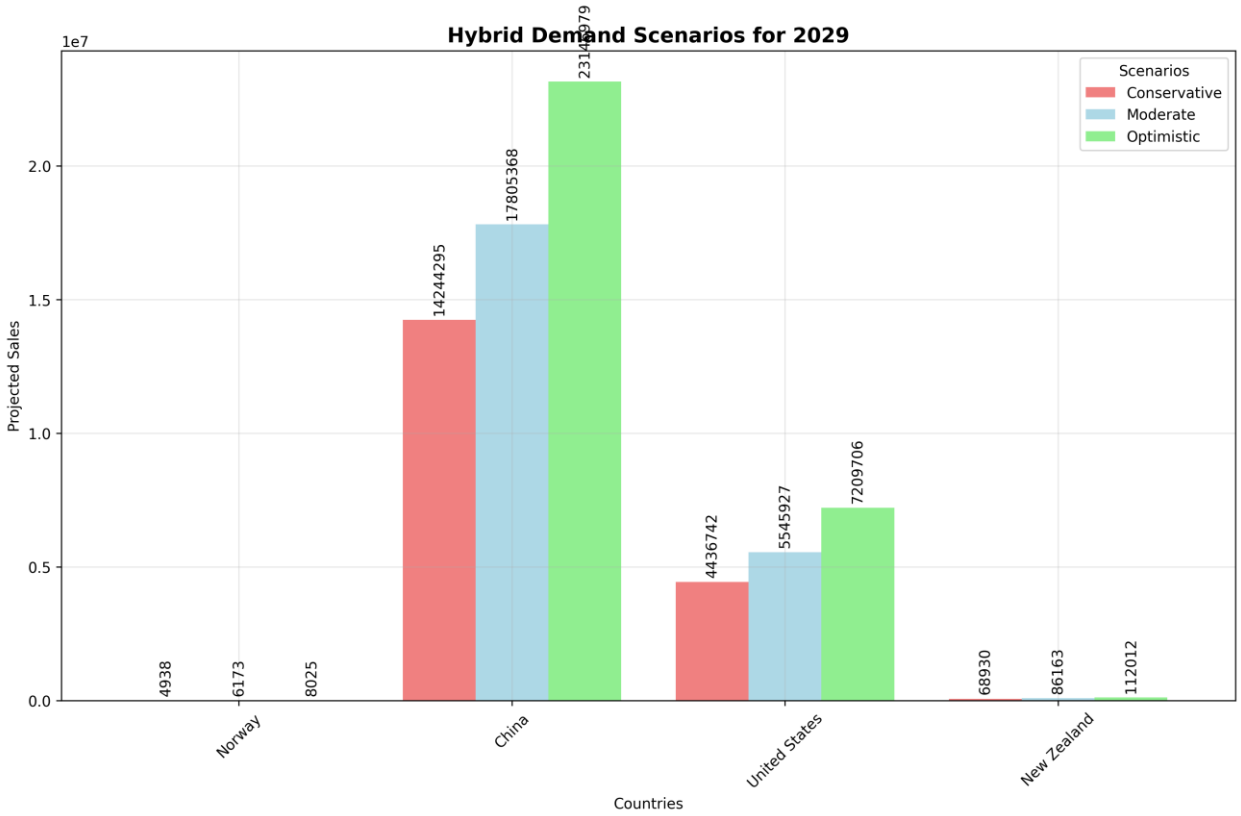
Price Elasticity by Country:



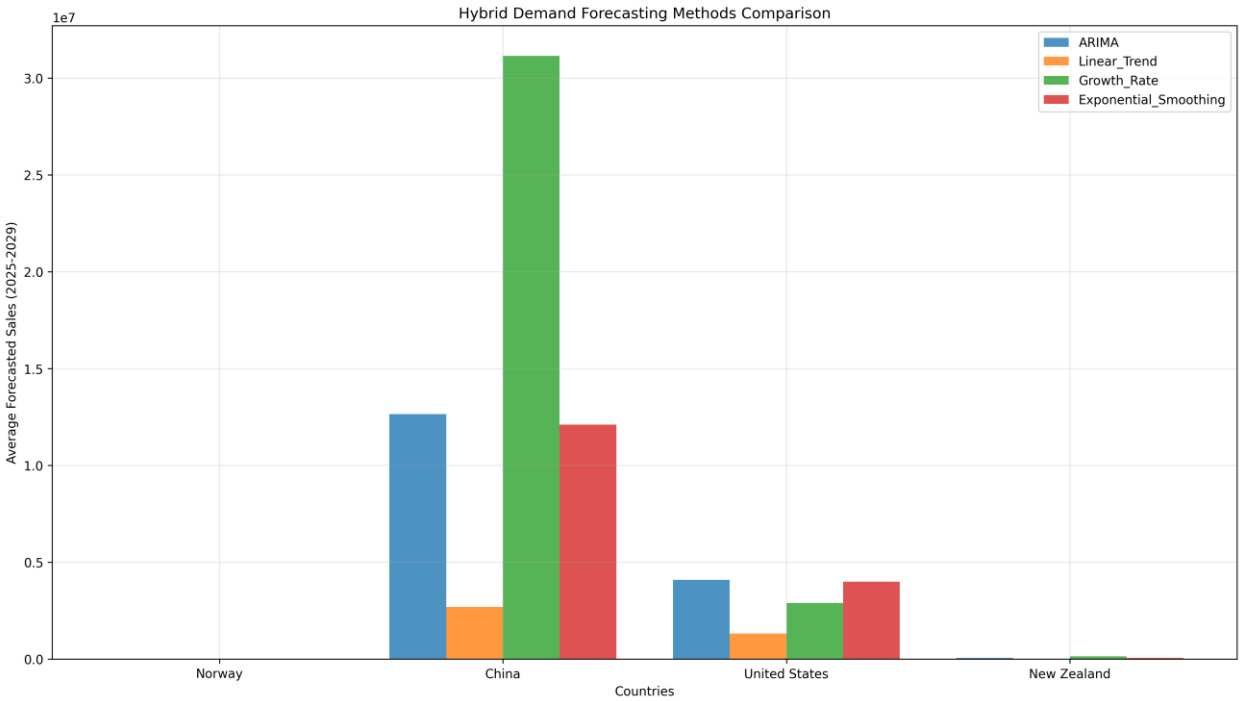
Demand Forecasting (Multiple Methods):



Scenario Analysis (2029 Projections):



Forecasting Methods Comparison:



3. ELECTRIC VEHICLE ANALYSIS

3.1 Descriptive Statistics

Overall Dataset Statistics:

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
Year	60	2017	4.36	2010	2013	2017	2021	2024
EV_Demand	60	1.62485e+06	3.65341e+06	24	5155.5	56226.5	896800	1.4174e+07
Fuel_Prices	60	1.28	0.45	0.51	0.88	1.24	1.67	2.12
Incentives	60	7075	5403.09	0	4000	7500	9750	15000
Technology	60	66313.4	195634	50	429.43	1672.92	14285.9	1.1112e+06
TCO	60	-1400	427.84	-2000	-1700	-1300	-1000	-1000
Behavioral_Factors	60	39.75	23.97	20	23.75	29.5	45.5	80

3.2 Correlation Analysis

Overall correlation between fuel prices and electric sales: -0.234

Lagged Correlations (Fuel Prices vs Sales):

| | Lag_0 | Lag_1 | Lag_2 | Lag_3 |

United States	0.356	0.341	0.11	-0.455
China	0.073	-0.051	-0.177	-0.398
Norway	0.39	0.134	-0.13	-0.315
New Zealand	0.638	0.531	0.05	-0.255

3.3 Regression Analysis & Price Elasticity

Country-Specific Price Elasticities:

Country	Simple Model	Multiple Model
United States	2.385	0.859
China	0.435	-0.245
Norway	2.837	-0.539
New Zealand	5.939	0.59

3.4 Granger Causality Test

Tests whether fuel prices 'Granger-cause' changes in vehicle sales:

Country	Min P-value	Significant (p<0.05)
United States	0.011	Yes*
China	0.6	No
Norway	0.247	No
New Zealand	0.381	No

*Indicates fuel prices Granger-cause vehicle sales at 5% significance level

3.5 Demand Forecasting Analysis

Multiple forecasting methods applied for 2025-2029 projections:

Average Annual Demand Forecast (2025-2029):

Country	ARIMA	Linear Trend	Growth Rate	Exp. Smoothing	Scenario Range
United States	1,208,126	1,109,415	6,882,872	1,556,000	966,500 - 1,570,563
China	16,634,622	17,275,458	133,711,050	17,212,328	13,307,697 - 21,625,008
Norway	129,447	156,990	482,853	143,531	103,557 - 168,281
New Zealand	-9,415	17,301	27,737	15,212	-7,532 - -12,239

Key Forecasting Insights:

• ARIMA models capture time series patterns and seasonality

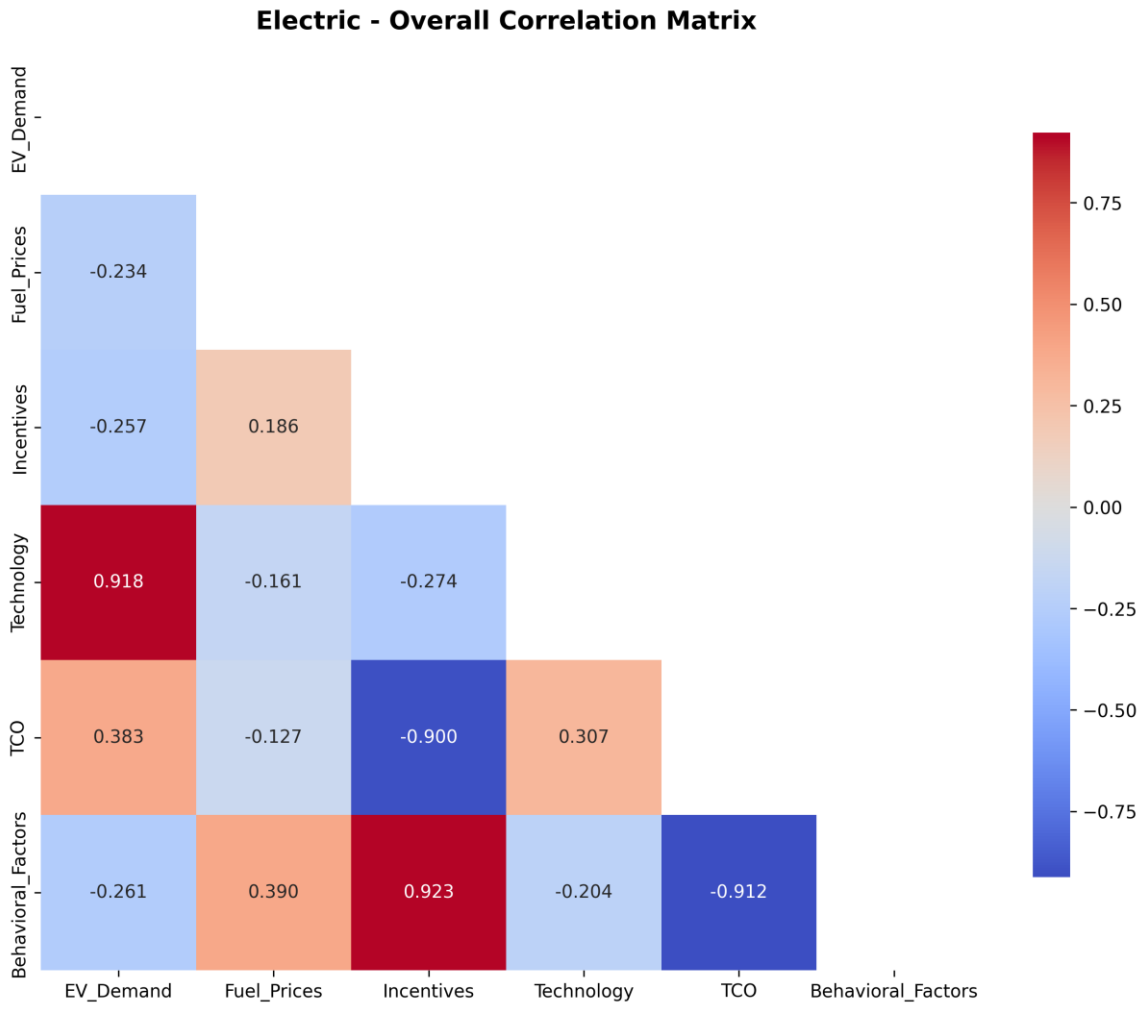
• Linear trends show long-term growth trajectories

• Growth rate models use historical CAGR for projections

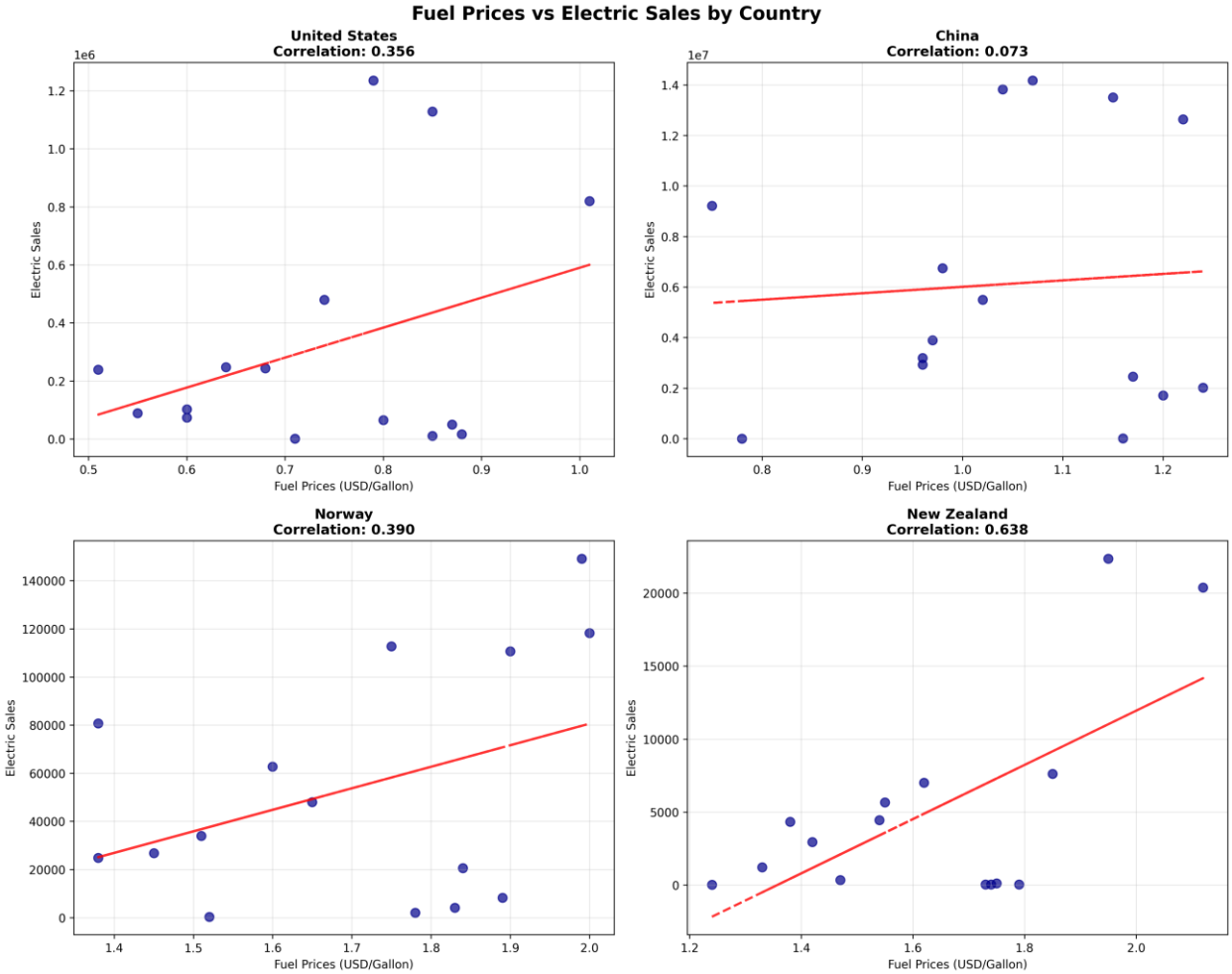
• Scenario analysis provides range of possible outcomes

3.6 Visualizations

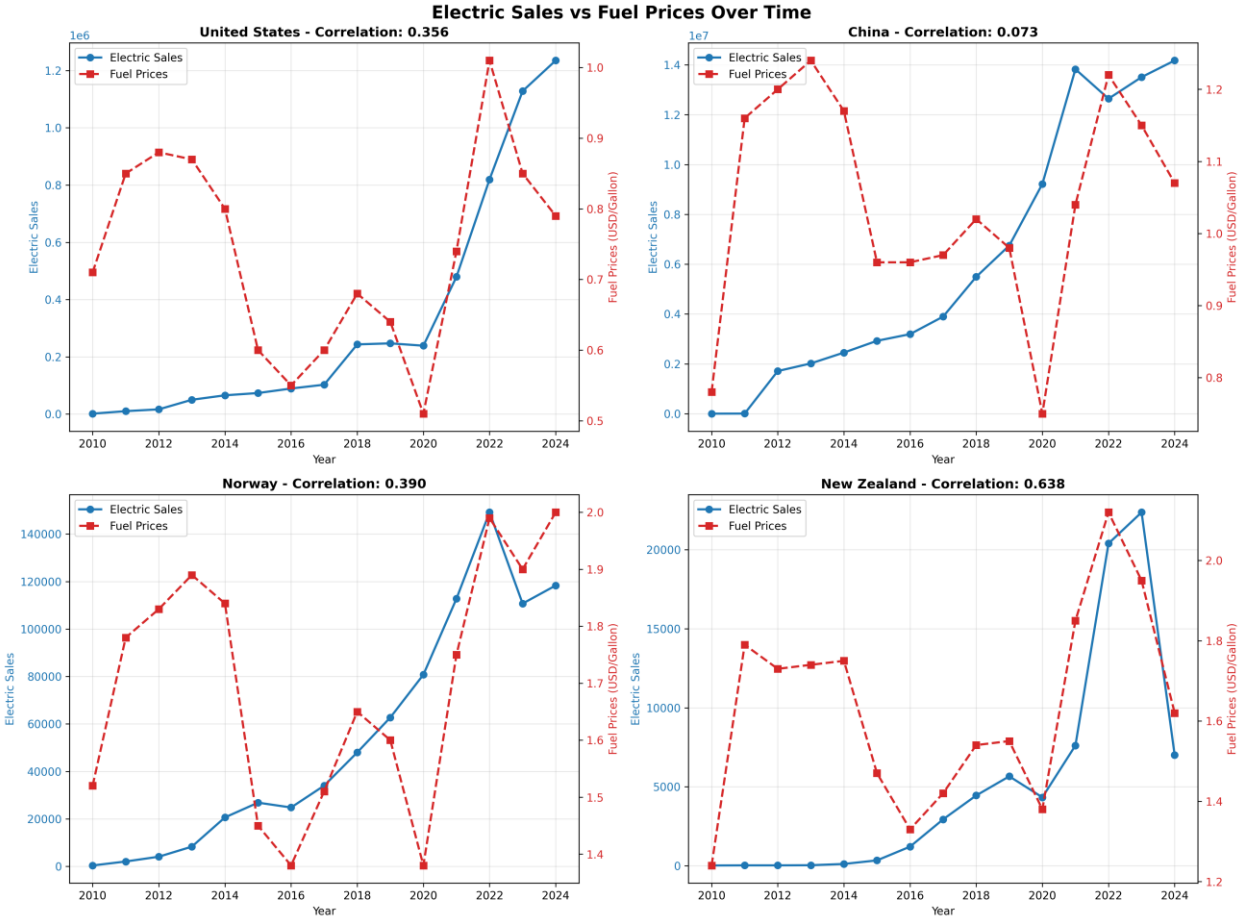
Correlation Matrix:



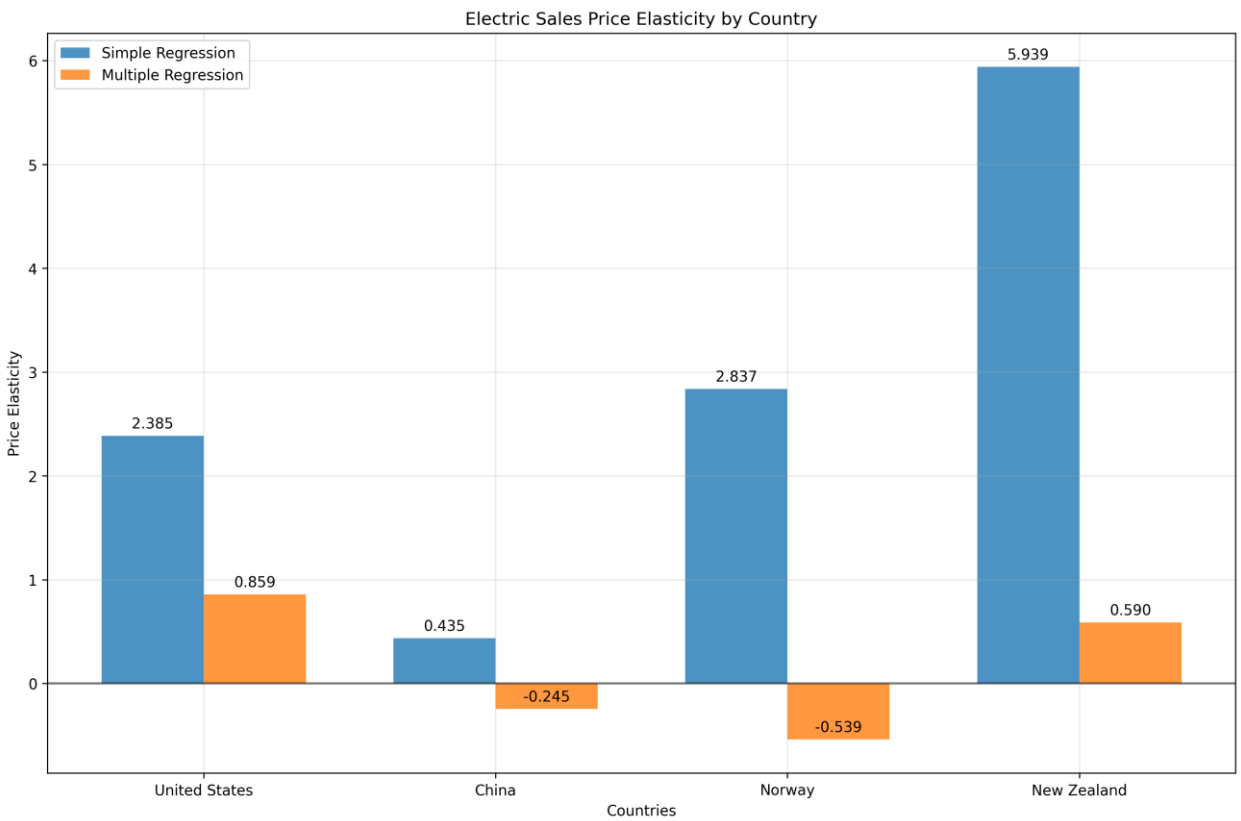
Fuel Prices vs Sales (Scatter Plot):



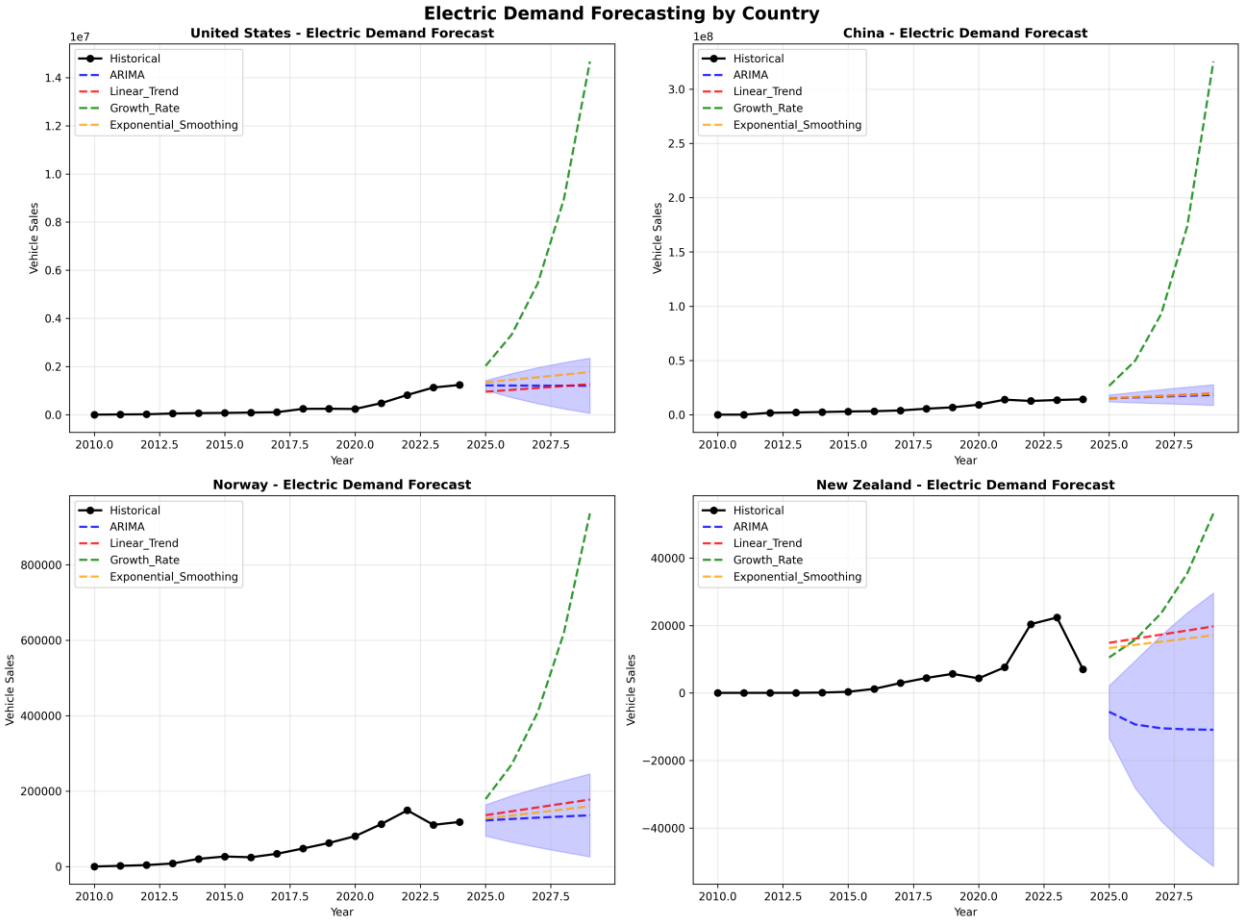
Time Series Analysis:



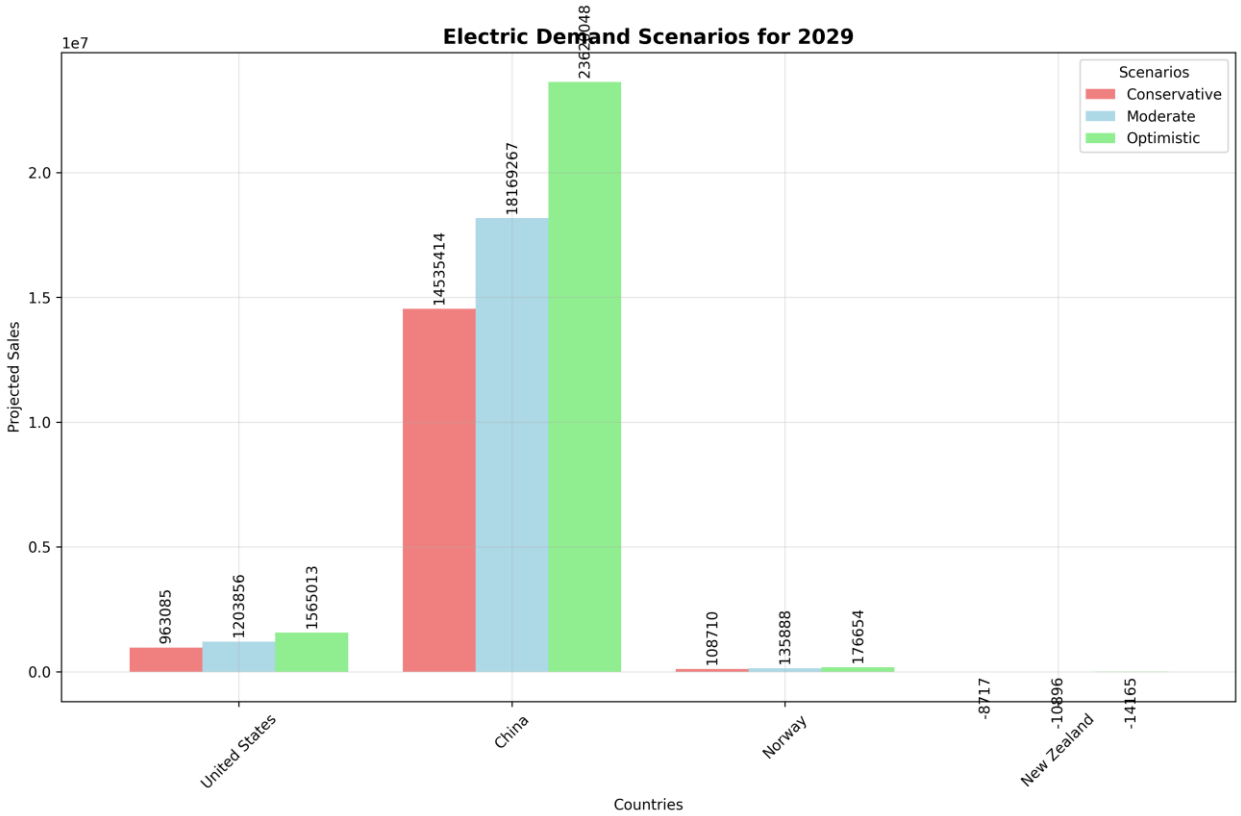
Price Elasticity by Country:



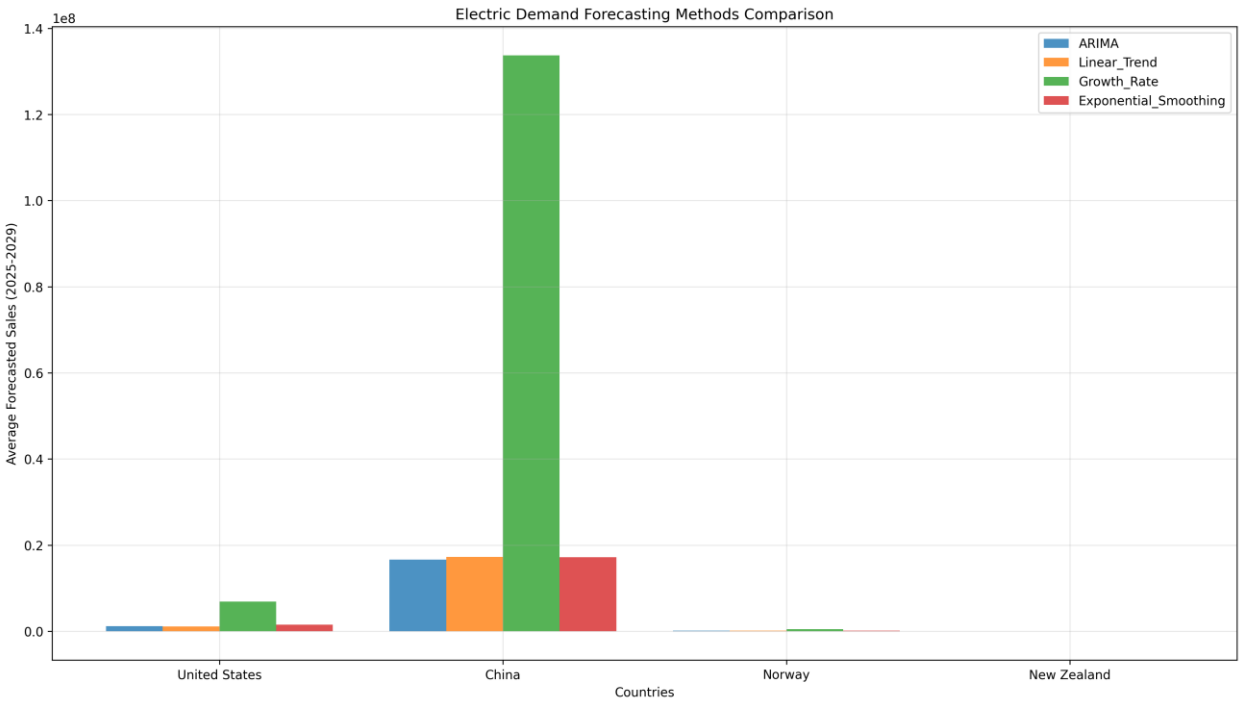
Demand Forecasting (Multiple Methods):



Scenario Analysis (2029 Projections):



Forecasting Methods Comparison:



4. COMPARATIVE ANALYSIS

FUEL PRICE SENSITIVITY COMPARISON:

Vehicle Type	Fuel Price Correlation	Interpretation
Hybrid Vehicles	-0.307	Moderate negative correlation
Electric Vehicles	-0.234	Weak negative correlation

DEMAND FORECAST COMPARISON (2025-2029 Average):

Country	Hybrid Vehicles (units/year)	Electric Vehicles (units/year)
United States	4,087,562	1,208,126
China	12,644,063	16,634,622
Norway	5,980	129,447
New Zealand	65,698	-9,415